

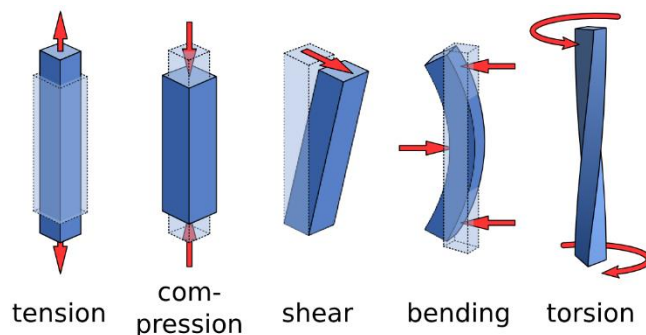
Elasticity

When studying matter, it is important to study its various properties. In the study of electrical properties, qualities like conductivity are important, while in the study of mechanical properties, elastic qualities such as hardness, strength, ductility, and rigidity are significant.

➤ Studying elasticity

When an object is subjected to external stress or compression, it undergoes deformation. This can be classified into three types.

1. One dimensional deformation
2. Two-dimensional deformation
3. Three-dimensional deformation



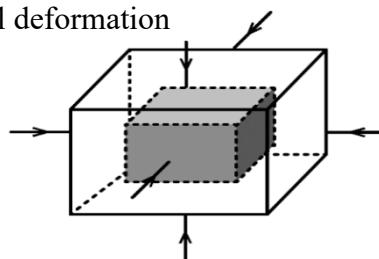
(a)

One dimensional deformation



(b)

Two-dimensional deformation



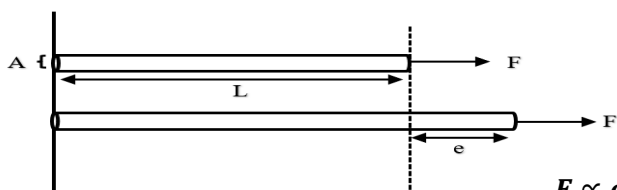
(c)

Three-dimensional deformation

- In this unit, we study the changes in shape that occur parallel to the direction of the force applied to an elastic object.

❖ Elastic rigid bodies

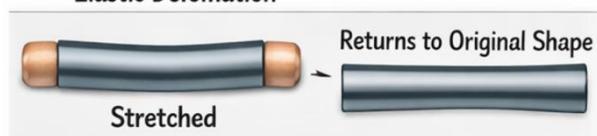
- Objects that change shape due to an external force but return to their original shape when the external force is removed are called **elastic objects**.
- By studying the one-dimensional deformation of an object by applying an external tension or compression force, the relationship between the external force and the change in the length of the object is presented as follows.



k = force constant

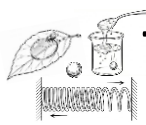
$$\begin{aligned} F &\propto e \\ F &= k e \\ \downarrow \quad \downarrow \downarrow \\ y &= m x \end{aligned}$$

Elastic Deformation



Plastic Deformation



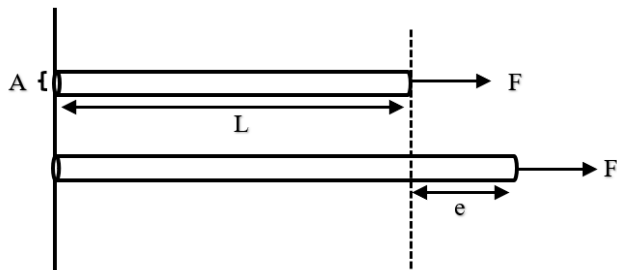


- The force constant (k) depends on the **material the object is made of, its initial length, and the cross-sectional area perpendicular to the force.**

❖ Stress (σ) and strain (ϵ)

Consider an elastic rod fixed at one end, with a force applied parallel to the rod's length and perpendicular to its cross-section. The increase in the rod's length under this force is called as extension. The decrease in the rod's length is called as compression.

The ratio of the tension or compression force to the perpendicular area is called as **stress** while the ratio of the extension (or compression) to the natural (original) length is called as **strain**.



A = cross sectional area

e = extension (units - m)

$\epsilon = \left(\frac{e}{L}\right)$ = strain (no units and dimensions)

strain = $\frac{\text{extension(compression)}}{\text{initial length}}$ • Pull – tensile strain

• Compression – compressive strain

$\sigma = \left(\frac{F}{A}\right)$ = stress (unit - N m^{-2} / Pa)

• Pull – tensile stress

• Compression- compressive

stress = $\frac{\text{elastic force}}{\text{initial length}}$

❖ Hooke's law for elastic objects

Within the proportional limit of an elastic object, the strain experienced by the object is directly proportional to the stress acting in the parallel direction.

$$\left(\frac{F}{A}\right) \propto \left(\frac{e}{L}\right)$$

$$\left(\frac{F}{A}\right) = Y \left(\frac{e}{L}\right)$$

Young's modulus
(Y or E)

$$\left(\frac{F}{A}\right) = Y \left(\frac{e}{L}\right)$$

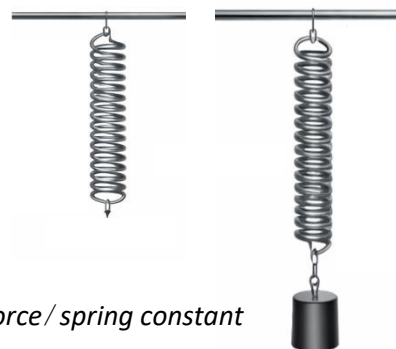
$$F = \frac{Y e A}{L}$$

$$F = \left(\frac{Y A}{L}\right) e$$

According to $F = ke$,

$$k = \left(\frac{Y A}{L}\right)$$

k = force / spring constant



When studying one dimensional deformations this is called as young's modulus while it's called as hardness modulus when studying torques and as Nikker's modulus when studying three dimensional deformations.

- The value depends solely on the material of the object.

Considering one dimensional deformations,

$$\left(\frac{F}{A}\right) = Y \left(\frac{e}{L}\right)$$

$$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{e}{L}\right)}$$

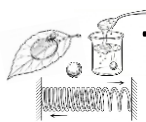
Unit of $Y \rightarrow \text{N m}^{-2}$ or Pa



Examples:



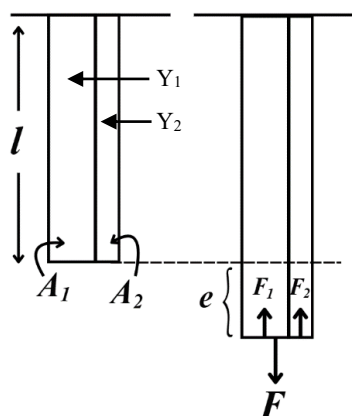
- 01) A 500g mass is hung from a wire 3m long with a cross-sectional area of 0.2 cm^2 . The length increases by 0.4cm. Calculate stress, strain and young's modulus of the wire with measurements mentioned above.



- 02) A hollow cylinder of 3m length and uniform cross-section is compressed by 0.025cm when a force of 3×10^5 N is applied. If the internal radius is 80% of the external radius and Young's modulus is $2.5 \times 10^8 \text{ N m}^{-2}$, find the external radius.
- 03) A wire with a cross-sectional area of 1 mm^2 and length 1m is fixed horizontally between two points. When a 40g mass is hung from the midpoint, it sags by 5mm. Calculate the Young's modulus of the metal.
- 04) A 1m long fiber is hung from a rigid support. When a 1kg mass is attached to its lower end, it extends by 1cm. If the mass is raised to the support and released, what is the maximum extension the fiber can reach?
- 05) A catapult is made using a rubber band of 42cm length and 3mm radius. The band is stretched by 20cm to launch a 20g object at a maximum speed of 20 ms^{-1} . Calculate the Young's modulus of the rubber.

❖ Parallel connection of two elastic strings

- Two rods/fibers with different Young's moduli and areas are connected such that their surfaces touch. When a force is applied to the combined string, both experience the same extension (e).
- In a manner suiting this, tension force divide among to strings.



$$\frac{F_1}{A_1} = Y_1 \frac{e}{l}$$

$$F_1 = \frac{A_1 Y_1 e}{l}$$

$$\frac{F_2}{A_2} = Y_2 \frac{e}{l}$$

$$F_2 = \frac{A_2 Y_2 e}{l}$$

$$F = F_1 + F_2$$

$$F = \frac{A_1 Y_1 e}{l} + \frac{A_2 Y_2 e}{l}$$

$$F = \frac{e}{l} (A_1 Y_1 + A_2 Y_2)$$

$$e = \left(\frac{Fl}{(A_1 Y_1 + A_2 Y_2)} \right)$$

❖ Series combination of two elastic strings

- Two different fibers connected end-to-end. The tension in both is the same.
- Total extension of the compound wire is equal to the separate sums of individual wires.

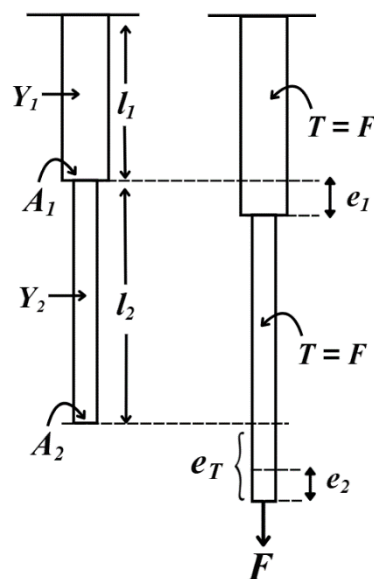
$$\frac{F}{A_1} = Y_1 \frac{e_1}{l_1}$$

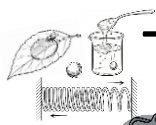
$$\frac{e_1}{F} = \frac{Fl_1}{A_1 Y_1}$$

$$e_1 = \frac{Fl_1}{A_1 Y_1}$$

$$e_T = e_1 + e_2$$

$$e_T = \frac{Fl_1}{A_1 Y_1} + \frac{Fl_2}{A_2 Y_2}$$



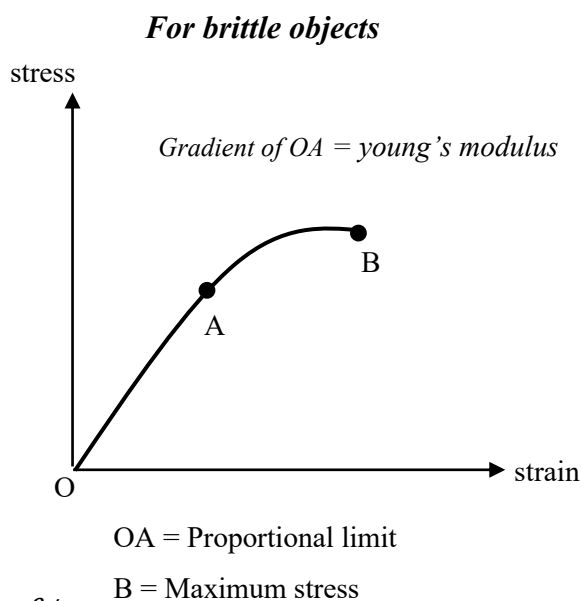
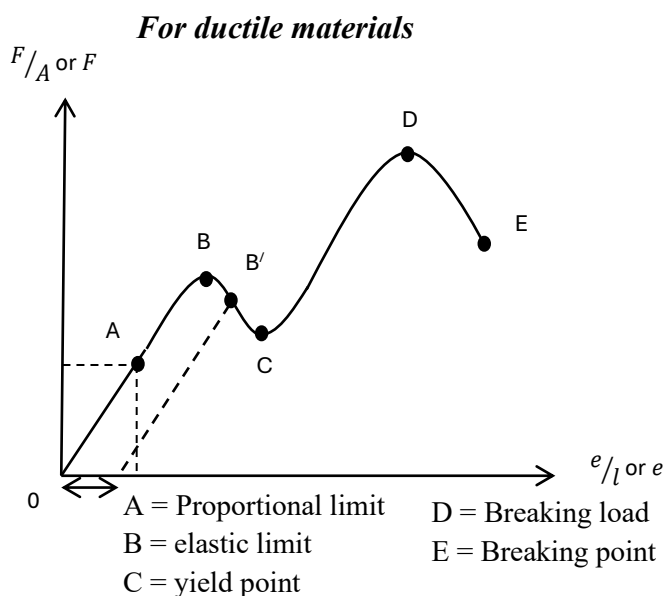


Example:

- 01) Two wires each having length of 0.5 cm and having areas in order 0.5cm^2 and 1.2cm^2 respectively are attached to each other to have a length 1m. Young's modulus and proportional limits are $2 \times 10^{11}\text{Nm}^{-2}$ and $2.5 \times 10^8\text{Nm}^{-2}$ in order. In a manner proportional limit doesn't exceed what is the maximum mass that can be hung? In this instance what is the total increment of compound wire? If these wires are arranged paralelly connecting the ends in a manner to have a length of 0.5 m what is the maximum mass the compound wire can hold such that proportional limit is not exceeded?

❖ Stress and strain variation of an elastic object

Stress against strain of an elastic object can be studied as below.



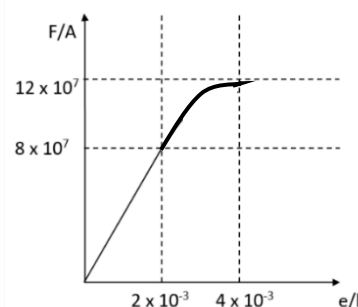
- In here A point is proportional limit. In OA zone, $F/A \propto e/l$.
- Until OB only one the object is released it comes to original position. When pulled to a point like B' finally some extension will remain.
- Maximum load that the object can hold is the breaking load. One it is given object easily reach breaking point and object breaks.
- Maximum stress that can be given to an object for it to be kept elastically is called as elastic limit.
- Consider:- **Once an object exceeds elastic limit once the external force is removed it's initial geometrical shape cannot be obtained.**

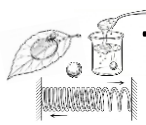


Example:

- 01) Shown below is extension- tension stress graph related to a copper wire. Initial length of the rod is 4m while cross sectional area is 10mm^2 .

- Using the graph find the young's modulus of wire.
- What is the stress at which proportional limit exceeds?
- When the wire breaks what is the tension of wire?
- Calculate the length when the wire breaks.





Strain energy stored in an elastic object in proportional limit

Elastic potential energy stored = Work done by mean force

$$E_P = \left(\frac{F_1 + F_2}{2} \right) \times S$$

$$\frac{F}{A} = E \frac{\Delta \ell}{\ell}$$

$$E_P = \left(\frac{0 + F_2}{2} \right) \Delta \ell$$

$$F = \left(\frac{EA}{\ell} \right) \Delta \ell$$

$$E_P = \frac{1}{2} \left(\frac{EA \Delta \ell}{\ell} \right) \Delta \ell$$

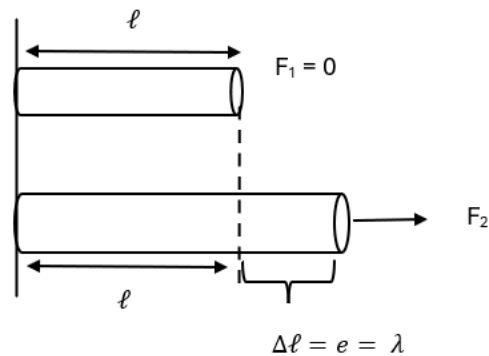
$$E_P = \frac{F_2}{2} \Delta \ell$$

$$E_P = \frac{F_2}{2} e$$

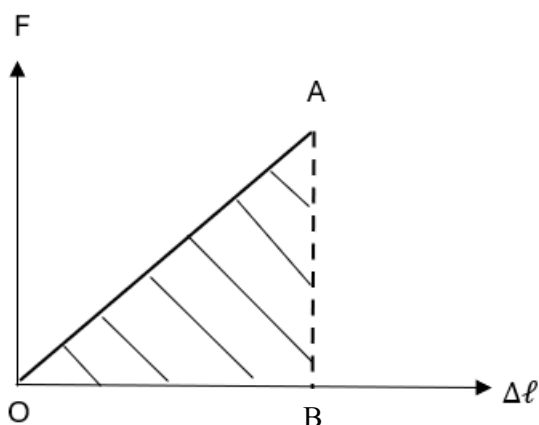
$$E_P = \frac{F_2}{2} x$$

$$E_P = \frac{1}{2} \left(\frac{EA}{\ell} \right) (\Delta \ell)^2$$

$$E_P = \frac{1}{2} kx^2$$



- On a force vs extension graph, the area under the graph represents strain energy.



Strain energy = OAB area

gradient = spring constant



➤ Strain energy in a unit volume (E_0)

Strain energy per unit volume = $\frac{\text{strain energy}}{\text{volume}}$

$$E_0 = \frac{\frac{1}{2} F \Delta \ell}{A \ell}$$

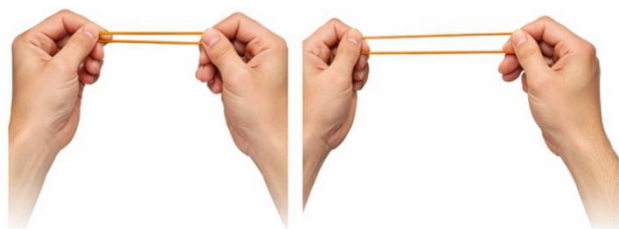
Strain energy per unit volume = $\frac{1}{2} \times \text{stress} \times \text{strain}$

$$E_0 = \frac{1}{2} \frac{F}{A} \frac{\Delta \ell}{\ell}$$

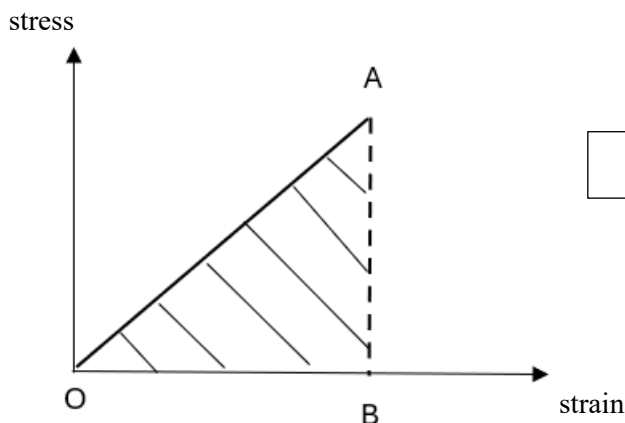
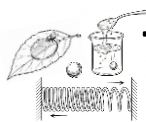
$$E_0 = \frac{1}{2} \sigma \varepsilon$$

$$E_0 = \frac{1}{2} \sigma \varepsilon$$

Unit of strain energy per unit volume = J m^{-3} or N m^{-2}



- Area covered under the graph of strain vs stress in proportional unit represents strain energy per unit volume.

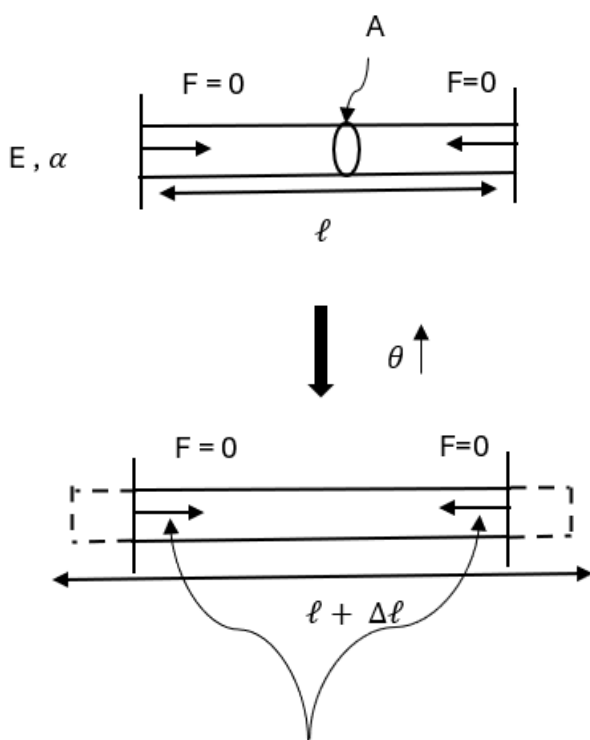


Strain energy per unit volume = OAB area



Strain forces due to stopping expansion or compression of a metal rod

Once expansion is stopped;



Tension force by wall on rod

When a metal rod of natural length l is placed between two fixed walls and the temperature is increased by θ , the rod attempts to expand. However, because the walls are rigid, they prevent this expansion by exerting a compressive force on the rod.

If this freely expanded increment in length,
 $\Delta l = l\alpha\theta$

In here imagine, a rod of length $(l + \Delta l)$ compressed to a length of Δl

Applying Hooke's law,

$$\frac{F}{A} = E \frac{\Delta l}{(l + \Delta l)}$$

$$\Delta l \ll l$$

$$l + \Delta l \approx l$$

$$\frac{F}{A} = E \frac{\Delta l}{l}$$

$$\frac{F}{A} = E \frac{l\alpha\theta}{l}$$

$$F = EA\alpha\theta$$

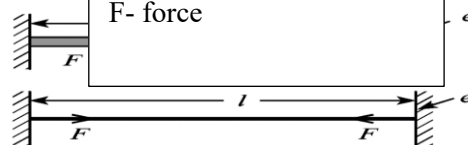
E- Young's modulus

A-Cross sectional area

α -Coefficient of linear expansion

θ - Temperature difference

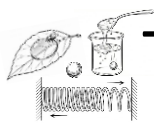
F- force



Example:

- 01) A steel rod with a cross-sectional area of 5 cm^2 at 25°C is heated to 65°C and then fixed between two rigid clamps. It is then cooled back down to 25°C . Calculate the tension force developed within the rod, assuming it remains within the proportional limit.

(Data given: Young's modulus of steel = $2 \times 10^{11} \text{ Nm}^{-2}$, coefficient of linear expansion = $1.2 \times 10^{-5} \text{ K}^{-1}$)



Elasticity

Short Essay Questions

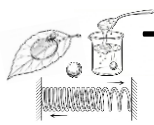
➤ Beginner Level

- (01) A steel wire used in a crane has a maximum allowable strain of 10^{-3} . If the radius of the wire is 4 mm and $E = 2 \times 10^{11} \text{ Nm}^{-2}$, find the maximum load it can lift. ($\pi = 3$)
(Answer : 9600 N)
- (02) Find the stress on a hollow cylinder with an outer radius of 2 m and inner radius of 1 m when a force of 10N is applied. ($\pi = 3$)
(Answer : 1.01 N m⁻²)
- (03) A platform of 500 kg is supported by 4 steel pillars, each 8 m high with a cross-section of 25 cm². Find the compression of each pillar ($E = 2 \times 10^{11} \text{ Nm}^{-2}$).
(Answer : $2 \times 10^{-5} \text{ m}$)
- (04) A string undergoes 1% strain. If the strain energy per unit volume is E, express the stress in terms of E.
(Answer : 200 E)
- (05) A 2 m wire is pulled with a 60 N force, resulting in a strain of 7.5×10^{-4} . Find the stored strain energy.
(Answer : $4.5 \times 10^{-2} \text{ J}$)
- (06) A rod ($E = 2 \times 10^{10} \text{ Nm}^{-2}$) is heated until it reaches 0.06% strain. Find the strain energy per unit volume. (Answer : $36 \times 10^2 \text{ J m}^{-3}$)

➤ Average Level

- (07) A copper wire has twice the radius and half the Young's Modulus of a steel wire. If they are connected in series and the copper wire extends by 1%, find the percentage extension of the steel wire.
(Answer : 2%)
- (08) A concrete pillar is reinforced with a steel rod in the center. The steel occupies 2.5% of the area, and its Young's Modulus is 10 times that of concrete. What percentage of the total load is carried by the steel?
(Answer : 20%)
- (09) A wire with area 1.14 m² ($E = 7 \times 10^8 \text{ Nm}^{-2}$) has a 2 m long object with a volume of $4 \times 10^{-4} \text{ m}^3$ attached. When submerged in a liquid, the extension decreases by 1 mm. Find the density of the liquid in kg m⁻³.
(Answer : 1×10^8)





- (10) As shown in the figure, a structure is prepared to hold a water tank of mass M using 4 identical metal rods, each with Young's modulus n , cross-sectional area A , and length l . If a liquid of density d is filled into the tank at a volume flow rate of V for the maximum possible time without exceeding the elastic limit of the rods, what is the change in the length of the rods compared to their initial length at that time?



(Answer : $\frac{l(dVt+M)g}{4An}$)

- (11) A wire with a breaking stress of $4.8 \times 10^7 \text{ N m}^{-2}$ has a length of 30 cm and a cross-sectional area of 1 mm^2 . A mass of 10 kg is attached to one end of the wire, and the other end is rotated so that the mass moves in a horizontal circle. Find the maximum angular velocity the particle can achieve.

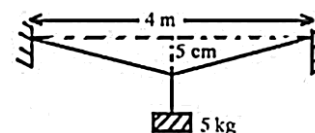
(Answer : 4 rad s^{-1})

➤ Master Level

- (12) One end of a light metal wire of length L and cross-sectional area A is fixed to a ceiling. The other end is attached to a light spring with a spring constant K . An object of mass m is hung from the free end of the spring. If the Young's modulus of the wire material is Y , find the total extension of the system.

(Answer : $mg \left(\frac{L}{YA} + \frac{1}{K} \right)$)

- (13) The length of a metal wire is 4 m and its cross-sectional area is 5 mm^2 . This wire is tied horizontally between two supports as shown in the figure below. When a load of 5 kg is hung from its midpoint, the midpoint of the wire drops by 5 cm from its initial position. What is the approximate value of the Young's modulus of the metal?



(Answer : $6.7 \times 10^{11} \text{ N m}^{-2}$)

- (14) The initial length of the elastic fiber of a catapult is 1 m. Its cross-sectional area is 1 mm^2 and its Young's modulus is $2 \times 10^8 \text{ N m}^{-2}$. When a pebble of mass 5 kg is placed in the fiber and it is stretched to an extension of 2 cm and released, find the velocity with which the pebble is launched.

(Answer : $12.6 \times 10^{-2} \text{ m s}^{-1}$)

MCQ

➤ Beginner Level

- (15) In a wire made of a material with Young's modulus $2 \times 10^{11} \text{ N m}^{-2}$, the strain at the proportional limit is 8×10^{-3} . If the tension in the wire at the proportional limit is $1.6 \times 10^4 \text{ N}$, the cross-sectional area of the wire is,

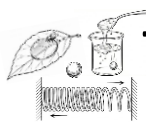
- (1) 2.5 mm^2 (2) 6.4 mm^2 (3) 10 mm^2 (4) 12.8 mm^2 (5) 100 mm^2

- (16) A uniform vertical steel wire is under tension due to a weight connected to its lower end. Which of the following would increase its extension? (Assume the Young's modulus of steel is greater than that of brass)

- A) Using a longer unstretched steel wire.
B) Using a steel wire with a larger cross-sectional area.
C) Using a brass wire of the same length and cross-sectional area.

- 1) Only A 2) Only C 3) Only A and B
4) Only B and C 5) All A, B and C





(17) Consider following statements.

- A) An object subjected to stress beyond the proportional limit will not return to its original length.
- B) An object subjected to stress beyond the elastic limit will not return to its original length.
- C) The area under a stress-strain graph represents strain energy density.

Which of the statements are true?

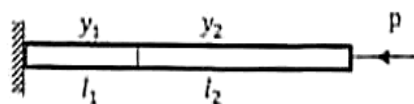
- (1) A only (2) B only (3) C only
- (4) A and B only (5) B and C only

(18) A person's weight W is supported by their two legs, each with a cross-sectional area S . If they stand on one leg, the stress on that leg becomes,

- (1) $S/2$ (2) S (3) $2S$ (4) $4S$ (5) $8S$

➤ Average Level

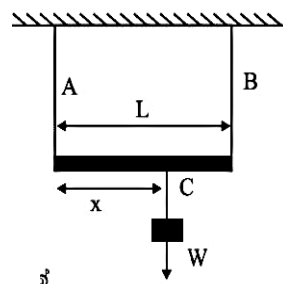
(19) Two rods of the same uniform cross-sectional area but with different lengths l_1 and l_2 are connected end-to-end and fixed between two rigid walls as shown. When the temperature of both rods is increased by the same amount, both rods undergo the same amount of compression. If their Young's Moduli are y_1 and y_2 and their coefficients of linear expansion are α_1 and α_2 , then,



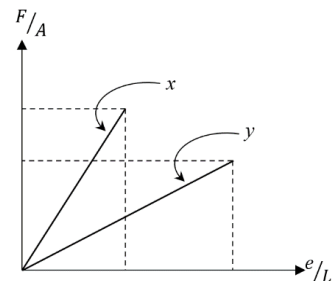
- (1) $l_1 l_2 = y_1 y_2$ (2) $l_1 y_1 = l_2 y_2$ (3) $l_1 y_2 = l_2 y_1$
- (4) $l_1 y_1^2 = l_2 y_2^2$ (5) $l_1 y_2^2 = l_2 y_1^2$

(20) A light horizontal rod of length L is suspended by two vertical wires A and B from a ceiling. The cross-sectional area of wire A is half that of wire B, while the Young's Modulus of wire A is twice that of wire B. A load W is hung at a distance x from wire A such that the stress in both wires is equal. The value of x is,

- (1) $L/3$ (2) $L/2$ (3) $2L/3$ (4) $3L/4$ (5) $3L/2$



(21) Two wires made of materials X and Y are suspended vertically and their lower ends are loaded with different weights. The graph shows the relationship between Stress (F/A) and Strain (e/L) for both wires. Both wires have the same length and they covered same area in the graph. Which of the following statements is/are true?



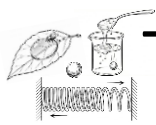
- (A) For a given load, the extension in wire Y is greater than the extension in wire X.
- (B) Young's Modulus of Y is less than Young's Modulus of X.
- (C) The cross-sectional area of Y is greater than the cross-sectional area of X.

- (1) A only (2) A and C only (3) B and C only
- (4) A,B and C all (5) A,B and C all incorrect

(22) A uniform steel wire has a length of 2 m and a cross-sectional area of 10^{-4} m^2 . Its Young's Modulus is $2 \times 10^{11} \text{ Nm}^{-2}$. The work done in stretching the wire by 0.1 mm is (assume the wire remains within the proportional limit),

- (1) $5 \times 10^{-1} \text{ J}$ (2) $5 \times 10^{-2} \text{ J}$ (3) $5 \times 10^{-3} \text{ J}$ (4) $5 \times 10^{-4} \text{ J}$ (5) $5 \times 10^{-5} \text{ J}$





➤ **Master Level**

- (23) The natural length of a rubber band is L_0 . When it is stretched to a length L , a small object of mass m is attached to it and released like a slingshot. If the cross-sectional area of the rubber band is A and Young's modulus is Y , find the maximum height (H) reached by the mass m . (Take the acceleration due to gravity as $g \text{ ms}^{-2}$).



(1) $H = \frac{Y \sin(\alpha)^2}{l_0} / 2g$

(2) $H = \frac{[2YA(l-l_0)^2 \sin \alpha]}{2gm}$

(3) $H = \sqrt{\frac{2YA}{l_0 m}} \cdot \frac{(l-l_0) \sin \alpha}{2g}$

(4) $H = \frac{YA(l-l_0)^2}{l \cdot mg}$

(5) $H = \frac{\left\{ \sqrt{\frac{2YA}{l_0 m}} (l-l_0) \sin \alpha \right\}^2}{2g}$

- (24) A rod of length l and cross-sectional area A is made of a material with Young's modulus Y and coefficient of linear expansion α . This rod is fixed between two rigid supports and its temperature is increased by θ . The tension force exerted by the rod on the supports is,

(1) $Ayl\alpha\theta$

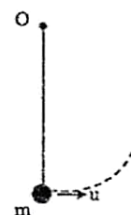
(2) $Ay^2\alpha\theta$

(3) $Ay\alpha\theta$

(4) $Ayl^2\alpha\theta$

(5) $Ay\alpha\theta/l$

- (25) A light elastic string of natural length l has one end attached to a fixed point O . A particle of mass m is attached to the other end and hangs in equilibrium. The spring constant of the string is k . If the mass is moving in a vertical circle, the minimum velocity u that must be given to the mass at its lowest position so that it just reaches the highest position is,



(1) $u = \sqrt{4gl - \frac{mg^2}{2k}}$

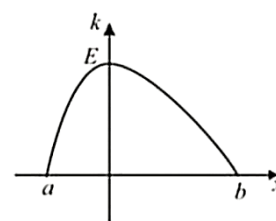
(2) $u = \sqrt{\frac{mg^2}{2k} - 2gl}$

(3) $u = \sqrt{2gl + \frac{mg^2}{k}}$

(4) $u = \sqrt{2gl + \frac{mg^2}{2k}}$

(5) $u = \sqrt{4gl + \frac{mg^2}{k}}$

- (26) A mass is attached to a light spring that obeys Hooke's Law. The system is set into oscillation such that its total mechanical energy is E . The graph shows how the kinetic energy k varies with the displacement x of the mass. The maximum force acts on the mass at,



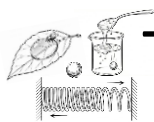
(1) At $x = 0$ only

(2) At $x = a$ only

(3) At $x = b$ only

(4) At $x = (a + b)/2$ only

(5) At $x = a$ only



Viscosity

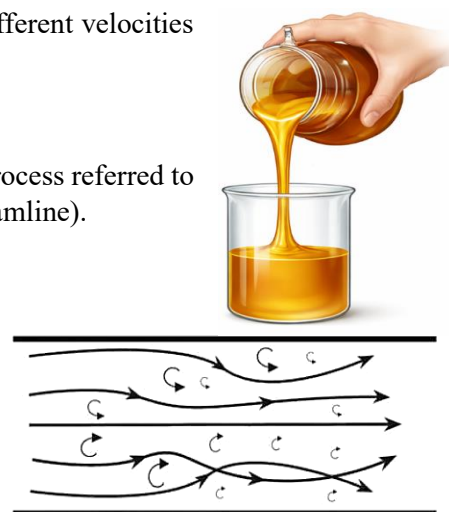
The internal resistance that arises between different layers of a fluid moving at different velocities relative to each other is known as **Viscous Force**.

Fluids with high viscosity exert higher frictional forces during flow.

Fluids always flow from a region of high pressure to a region of low pressure, a process referred to as Fluid Flow. Fluid flow can be categorized as either Turbulent or Laminar (Streamline).

Turbulent flow

Flow that is disorganized is called turbulent flow. In such flow, different particles passing through a specific point may have different velocities and move in different directions. Turbulent flow usually occurs when a fluid flows under a very high pressure difference.



Streamline (laminar) flow

Flow is considered streamline if all particles passing through a specific point in the fluid have the same velocity magnitude and direction.

The path taken by a particle moving in a streamline flow is represented by Streamlines. Streamlines never intersect each other. The direction of the tangent drawn to a streamline at any point indicates the direction of velocity of the particle passing through the point.



Steady flow

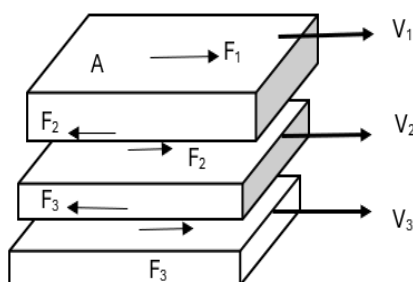
Steady flow occurs when a fluid flows such that all particles passing through the same streamline have the same velocity. Steady flow can be visualized as the fluid moving in distinct layers. Newton presented a relationship for the tangential stress between these layers in a steadily flowing viscous fluid.

Viscous nature

When fluid layers move relative to each other, a frictional force (viscous force) exists between them. The property of a fluid that causes this internal friction is called Viscosity. This property arises due to the intermolecular forces between the fluid molecules.

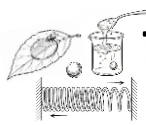


How the velocity varies between fluid layers in a steady flow

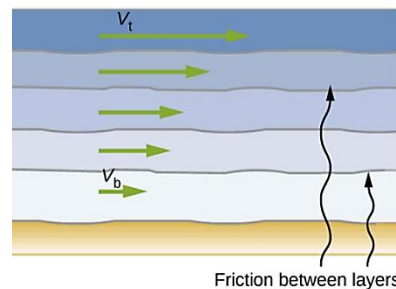
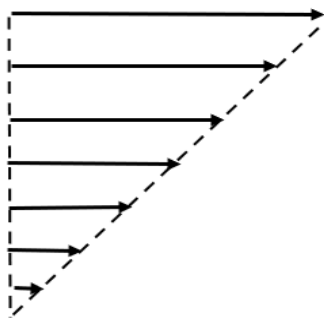


Consider a fluid layer with a surface area A . Suppose a viscous force F_1 acts on the layer immediately above it. This layer then begins to move with a velocity V_1 . Because of this movement, a viscous force F_2 is exerted on the layer directly below it, causing it to move with a velocity V_2 . Similarly, the layer below that moves with velocity V_3 due to the viscous force F_3 exerted by the layer above it.

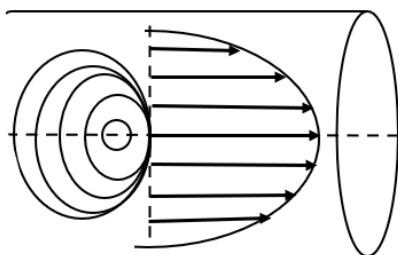




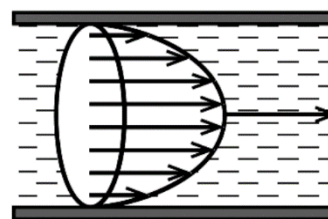
In this manner, the velocity is transmitted from layer to layer through the viscous friction force. The diagram below illustrates how the velocity of fluid layers changes when a fluid flows steadily through a vertical plane with a constant cross-section. The velocity of the fluid layers decreases as the distance from the moving surface increases.



Velocity Distribution of Fluid Flowing Steadily Through a Cylindrical Tube



When a fluid flows through a tube, the layer in contact with the axis of the tube moves with the maximum velocity. As you move from the axis toward the walls of the tube, the velocity of the fluid layers gradually decreases. The velocity of the fluid layer in direct contact with the tube wall is zero.

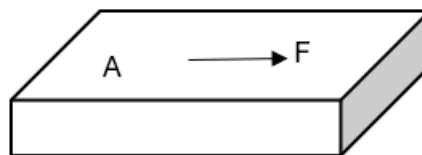


Tangential stress

The viscous friction force acting tangentially on a unit surface area of the fluid layers is called Tangential Stress.

$$\text{Tangential stress} = \frac{\text{Viscous friction force}}{\text{Area}}$$

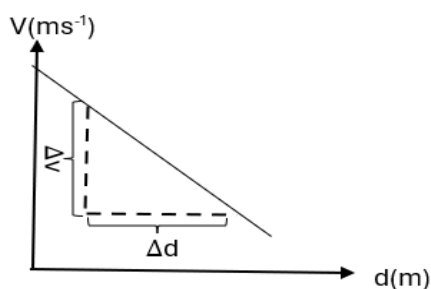
$$P = \frac{F}{A}$$



Unit of tangential stress $\longrightarrow$ N m^{-2} or Pa

Velocity gradient

The rate of change of velocity with respect to the perpendicular unit distance to the fluid layers is known as the Velocity Gradient.

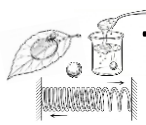


$$\text{Velocity gradient} = \frac{\Delta v}{\Delta d}$$

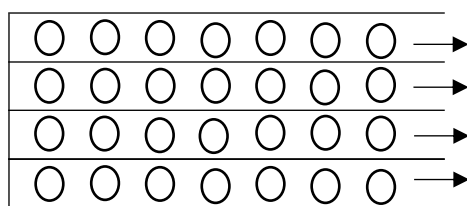
$$\text{Unit} = \frac{\text{ms}^{-1}}{\text{m}}$$

$$= \text{s}^{-1}$$

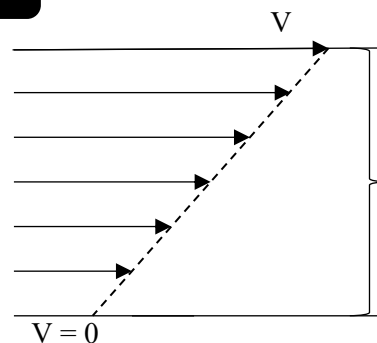




➤ Newton's law on viscosity



streamlined



f = liquid friction
(viscous force)

"For a fluid flowing in a steady (laminar) state, the tangential viscous stress between any two layers is directly proportional to the velocity gradient between those layers"

Tangential stress

$$\left(\frac{v}{d}\right) = \frac{(v_1 - v_2)}{d} \quad \eta = \text{coefficient of viscosity}$$

$$= \frac{\Delta v}{d}$$

$$\frac{f}{A} \propto \frac{\Delta v}{d}$$

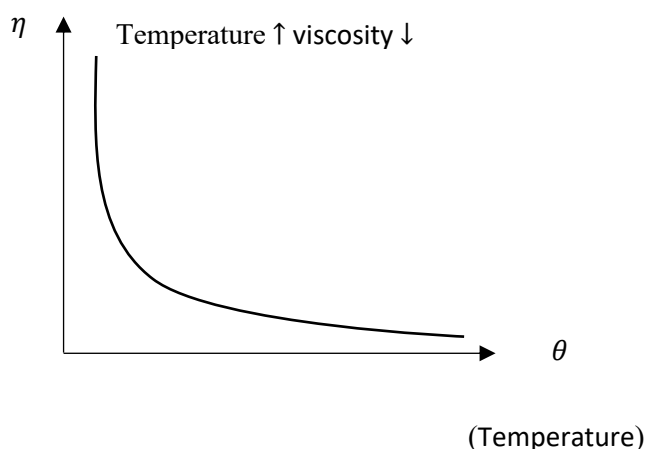
$$\frac{f}{A} = K \frac{\Delta v}{d}$$

$$\frac{f}{A} = \eta \left(\frac{\Delta v}{d}\right)$$

$$= \frac{\left(\frac{f}{A}\right)}{\left(\frac{v}{d}\right)}$$

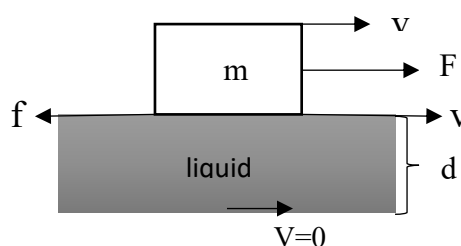
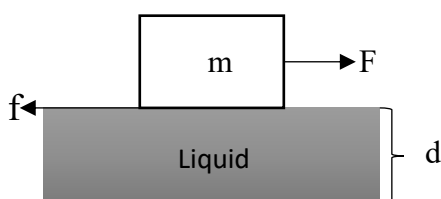
$$= \frac{Pa}{\left(\frac{ms^{-1}}{m}\right)}$$

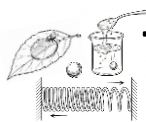
$$= Pa \cdot s$$



➤ Attaining Terminal Velocity Under a Constant External Force

Consider a mass m placed on a horizontal surface covered with a layer of viscous fluid of thickness d . A constant horizontal external force F is applied to the mass.

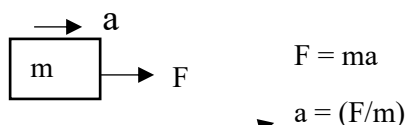




At the start ($V = 0$)

$$\frac{f}{A} = \eta \frac{v}{d}$$

Since $V = 0$, $f = 0$



Initial acceleration
(When acceleration is maximum)

Terminal velocity instance (v_0)

At a certain moment,

$$f = F$$

$$a = \frac{F-f}{m}$$

$$a = \frac{0}{m}$$

$$a = 0 \text{ ms}^{-2}$$

When $f = F$,

$$F = A\eta \frac{v_0}{d}$$

$$v_0 = \frac{Fd}{A\eta}$$

After a short time (When velocity is V)

$$\frac{f}{A} = \eta \frac{v}{d}$$

$$f = A\eta \left(\frac{v}{d}\right)$$

Viscous force

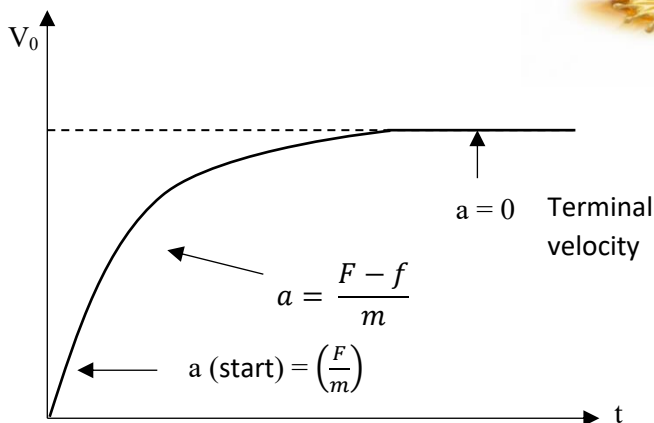
Acceleration of object

$$F = ma$$

$$F - f = ma$$

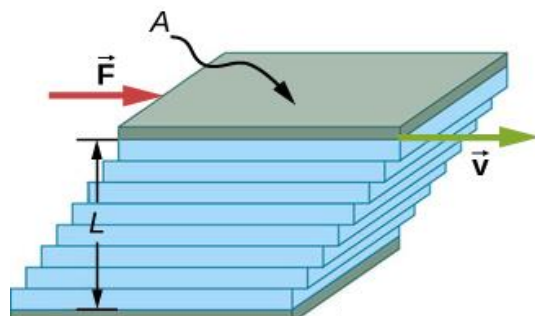
$$\downarrow a = \left(\frac{F-f}{m}\right)$$

$$v \uparrow \rightarrow f \uparrow \rightarrow a \downarrow$$



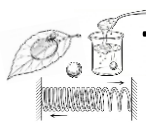
Examples:

- 01) A horizontal plate with a surface area of 200 cm^2 is placed on a layer of oil that is 1 mm thick. Calculate the force required to move the plate at a constant velocity of 10 ms^{-1} across the surface. The coefficient of viscosity for the oil is $1.1 \times 10^{-3} \text{ Pa s}$.



- 02) A wooden block of mass 5 kg is placed on a horizontal surface covered with a layer of oil with a coefficient of viscosity of $2 \times 10^{-2} \text{ Pa s}$. The thickness of the oil layer is 5 mm and the contact area of the block is 50 cm^2 .
- What is the force required to move the object forward at a constant velocity of 10 m s^{-1} ?
 - If a constant force of 10 N is applied to the object, find the terminal velocity reached. Draw a graph showing the variation of velocity with time for this motion.
 - If the surface is tilted at an angle of 30° to the horizontal, find the terminal velocity and the initial acceleration of the block as it begins to move.

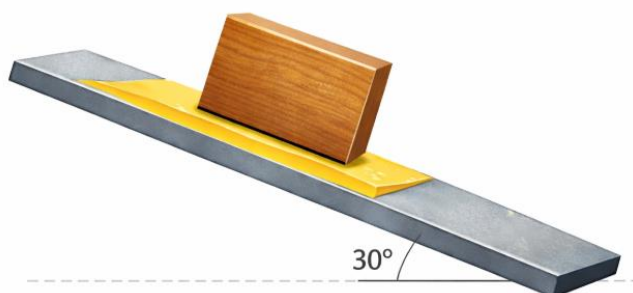




03) A wooden block of mass 1 kg and base area 50 cm^2 is placed on a plane inclined at 30° to the horizontal. A 2 mm thick layer of oil is between the block and the plane. The block slides down the plane at a terminal velocity of 10 m s^{-1} . Calculate the coefficient of viscosity of the oil used. What is the acceleration of the block at the moment it begins its motion?

04) A liquid flows through a horizontal surface with a internal radius of 5 cm. The velocity gradient of the surface layer of this liquid is 1 s^{-1} . Calculate the viscous force acting on a flat plate with a surface area of 20 cm^2 that is placed on the surface of this moving liquid. (The coefficient of viscosity of the liquid is $1.1 \times 10^{-3} \text{ Pa s}$).

05) A layer of oil 1 mm thick is on a plane inclined at 30° to the horizontal. A wooden block of mass 0.1 kg and base area $4 \times 10^{-2} \text{ m}^2$ is placed on it. The coefficient of viscosity of the oil is $1 \times 10^{-2} \text{ N s m}^{-2}$.

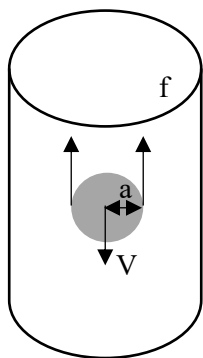


- The block begins to move down the plane. At a moment when its velocity is v , what is the viscous friction force acting against its motion?
- Derive an expression for the acceleration of the block at the moment its velocity is v . (Take $g = 10 \text{ m s}^{-2}$)
- What is the maximum acceleration of the block? At what point does this occur?
- When the block's acceleration becomes zero, what is its velocity?
- What is the terminal velocity reached by the block?
- Draw graphs for acceleration vs. Time, Velocity vs. Time, and Displacement vs. Time for the block's motion.,

(Ans :- (a) $0.4v$ (b) $5 - 4v$ (c) 5 m s^{-2} , obtains when the motion starts (d) zero, 1.25 m s^{-1} (e) 1.25 m s^{-1})

➤ Stroke's equation

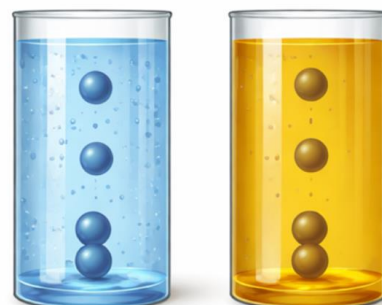
When a small spherical object moves through a viscous fluid, a resistive force (viscous force) acts against the direction of motion. Stokes found that this force is directly proportional to the radius of the sphere, its velocity, and the viscosity of the fluid.

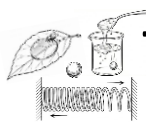


$a = \text{radius of sphere}$

$$\begin{aligned} f &\propto a \\ f &\propto \eta \\ f &\propto v \\ f &\propto a\eta v \\ f &= (k)a\eta v \\ &\downarrow \\ f &= (6\pi)a\eta v \end{aligned}$$

$$\boxed{\uparrow f = 6\pi\eta a v \uparrow}$$

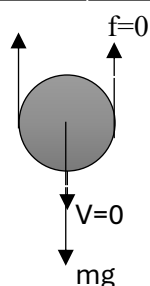




➤ Necessary conditions for Stoke's law to apply

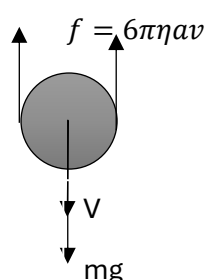
- The object must be perfectly spherical.
- The fluid medium through which the object moves must be infinite (the container walls should not affect the flow).
- The medium must be homogeneous and continuous.
- The velocity of the object must be small enough to maintain streamline flow around it.
- There should be no slipping between the object's surface and the fluid layer in contact with it.

➤ Motion of a spherical object falling through air



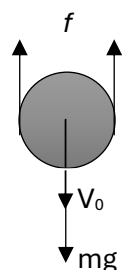
Initially when the motion starts

$$\begin{aligned} \downarrow F &= ma \\ mg &= ma \\ a' &= g \downarrow \end{aligned}$$



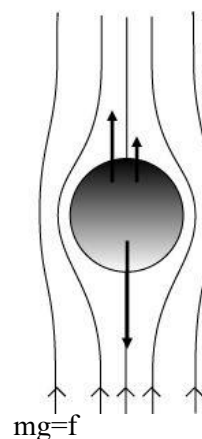
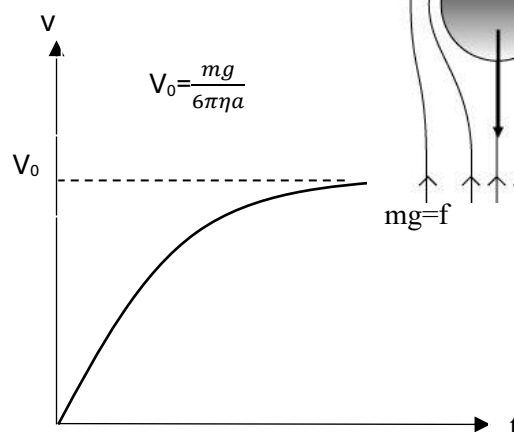
Reasonable instance

$$\begin{aligned} \downarrow F &= ma \\ (mg - F) &= ma' \\ \downarrow a' &= \frac{(mg - f \uparrow)}{m} \\ v \uparrow \Rightarrow f \uparrow \Rightarrow a' \downarrow \end{aligned}$$



At a certain instance

$$\begin{aligned} f &= mg \\ a' &= 0 \\ 6\pi\eta av_0 &= mg \\ v_0 &= \frac{mg}{6\pi\eta a} \end{aligned}$$



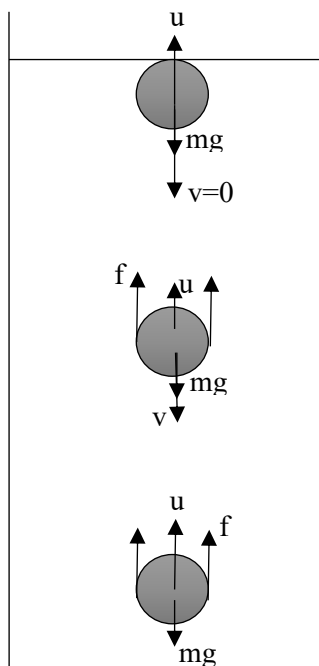
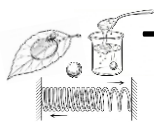
a' = acceleration
 a = radius of sphere

➤ Motion of a spherical object through a liquid

When an object's density is higher than the density of the liquid, it moves downward through the liquid. As the velocity increases, the viscous force also increases, and so that acceleration of the sphere decreases.

With velocity due to increment of viscous force at a certain instance resultant force becomes zero and acceleration becomes zero and then moves with constant velocity. This is called as **terminal velocity**. Consider a sphere falling down starting from rest.





Initially

$$\downarrow F = ma$$

$$mg - u = ma$$

$$a = \frac{(mg - u)}{m}$$

When moving with V velocity

$$\downarrow F = ma$$

$$mg - u - f = ma$$

$$a = \frac{(mg - u - f \uparrow)}{m}$$

At a certain instance

$$f + u = mg$$

Then $a=0$

$\therefore$ it reaches terminal velocity



$$f + u = mg$$

$$6\pi\eta av_0 + vdg = (vd)g$$

$$6\pi\eta av_0 = vdg - v\rho g$$

$$6\pi\eta av_0 = vg(d - \rho)$$

$$6\pi\eta av_0 = \frac{4}{3}\pi a^3 g(d - \rho)$$

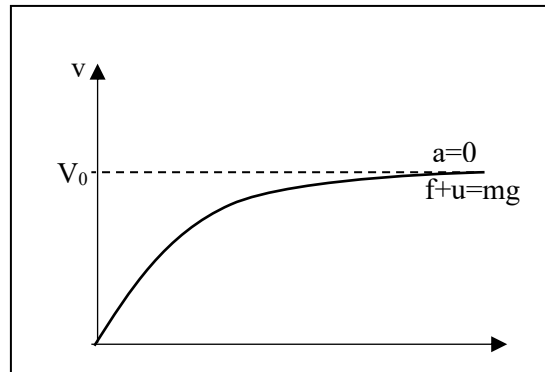
$$v_0 = \frac{4}{18} \frac{a^2}{\eta} (d - \rho)g$$

$$v_0 = \frac{2}{9} \frac{a^2}{\eta} (d - \rho)g$$

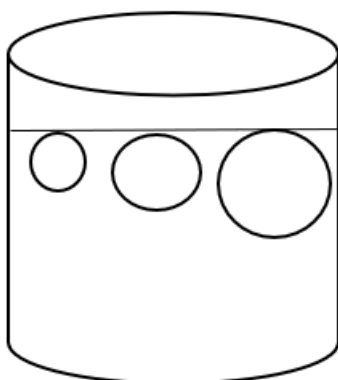
d = density of object

ρ = density of liquid

v = volume of sphere



Two spheres made of same material but with two different radii freed in same liquid, sphere with largest radii reaches the highest terminal velocity.



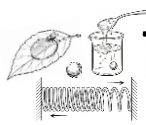
$$v \propto r^2$$

$$\frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$$

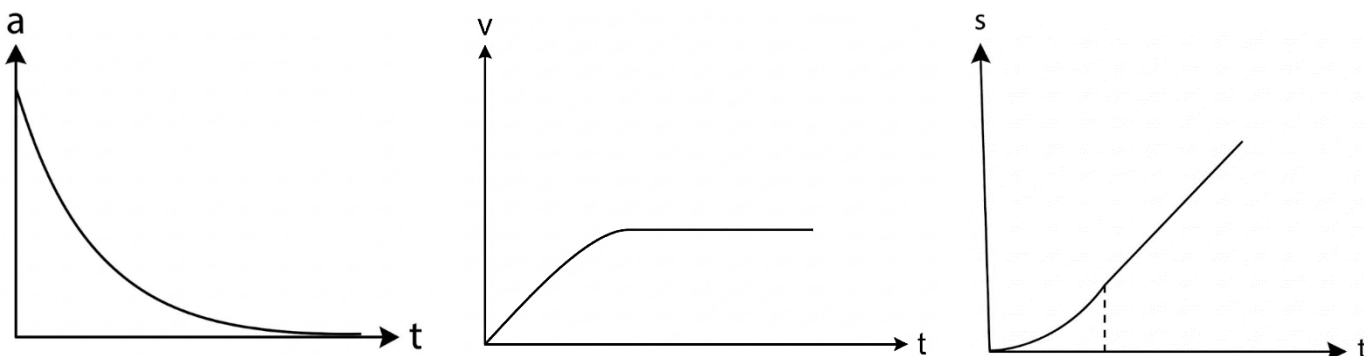
If the density of the spherical object is lesser than density of fluid, sphere moves up through the fluid. Then the terminal velocity obtained by object is

$$v = \frac{2r^2}{9\eta} (\rho - d)g$$

$$\rho > d$$



$a-t$, $v-t$, $s-t$ graphs relevant to the motion of the sphere



Eg:-

01) A spherical object with a density of 0.1 g cm^{-3} and a radius of 5 mm is released from the bottom of a deep tank of liquid. The viscosity of the liquid is 0.01 N s m^{-2} and its density is 1 g cm^{-3} .

- Calculate the net upward force acting on the object at the moment it is released.
- Calculate the acceleration of the object at that initial moment.
- What is the terminal velocity of the sphere when it reaches the surface?

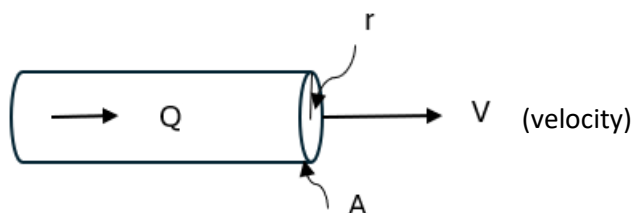
02) Small sand drops are dropped onto the surface of a deep tank filled with water to a height of 10 cm. The diameter of the ball is 0.1 mm and it is assumed to reach terminal velocity almost instantly. Calculate the time taken to reach the bottom. (Take coefficient of viscosity for water as $1.01 \times 10^{-3} \text{ N s m}^{-2}$, density of water as 1000 kg m^{-3} , and density of the ball as 2200 kg m^{-3}).

03) A lead sphere of radius 1 mm falls through a tank containing glycerin. Find the velocity of the sphere at the moment its acceleration is exactly half of its initial acceleration. Also, find the terminal velocity. (Density of lead = 8.50 g cm^{-3} , density of glycerin = 1.23 g cm^{-3} , and coefficient of viscosity for glycerin = 0.5 N s m^{-2}).

Flow of a viscous fluid through a horizontal tube

1. Volume flow rate (Q)

The volume of fluid flowing through a cross-section of a tube per unit time is called the "Volume Flow Rate."



If V liquid volume flows in a t time,

$$Q = \frac{V}{t}$$

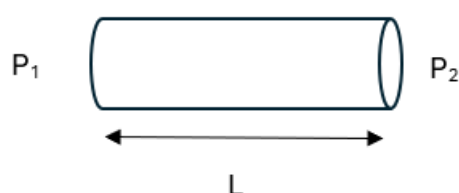
$$\text{unit}(Q) \rightarrow \text{m}^3 \text{ s}^{-1}$$

$$Q = Av$$

$$Q = \pi r^2 v$$

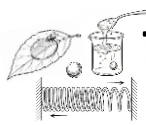
2. Pressure gradient

The change in pressure per unit length along the direction of fluid flow is called the "Pressure Gradient."



$$\text{Pressure gradient} = \frac{\Delta P}{\Delta L}$$

$$\text{Pressure gradient} = \frac{P_1 - P_2}{\Delta L}$$



$$\text{Unit of pressure gradient} = \frac{Nm^{-2}}{m}$$

$$= Nm^{-3}$$

$$[\text{Pressure gradient}] = MLT^{-2} \times L^{-3}$$

$$= ML^{-2}T^{-2}$$

Rate of work to be done to move a viscous fluid through a tube (P_0)

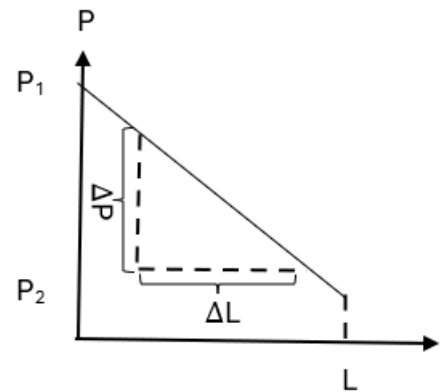
$$P_0 = F \cdot V$$

$$F = (\Delta P) \cdot A$$

$$P_0 = (\Delta P) A \cdot V$$

$$Q = AV$$

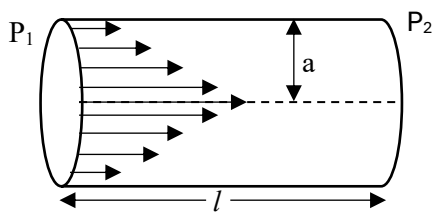
$$P_0 = (\Delta P) Q$$



➤ Poiseuille's Equation for the Flow of a Viscous Fluid

To flow a viscous fluid through a tube, a pressure difference must be maintained between the two ends. Fluid flows from a region of **high pressure to a region of low pressure**.

Poiseuille derived an equation for the volume flow rate (Q) of a viscous fluid through a cylindrical tube under a steady pressure gradient. The flow depends on the tube's length, its radius, the viscosity of the fluid, and the pressure difference.



$$P_1 > P_2$$

Fluid flows from higher pressure to lower pressure.

Poiseuille's equation;

$$Q = \frac{\pi a^4}{8\eta} \frac{\Delta p}{l}$$

a = radius

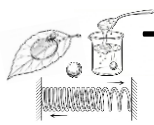
η = coefficient of viscosity

$\left(\frac{\Delta p}{l}\right)$ = pressure gradient

Q = rate of volume flow

➤ Conditions for Poiseuille's equation to be valid

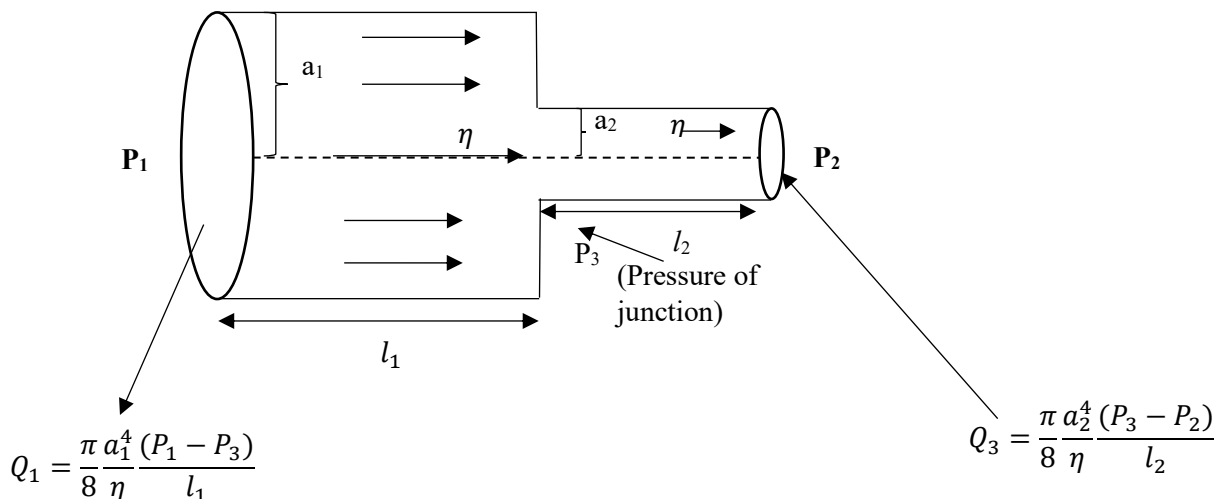
- The flow must be steady (laminar).
- The tube must be horizontal.
- The fluid must be incompressible and have constant viscosity.
- The fluid layer in contact with the tube wall must be at rest (zero velocity).
- This equation is most accurate for "narrow" tubes (capillary tubes).



➤ Flow of viscous fluid through a composite tube system

01) Tubes connected in series

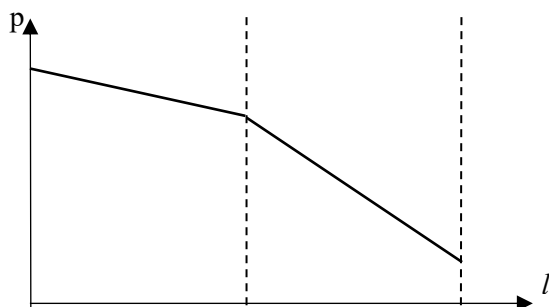
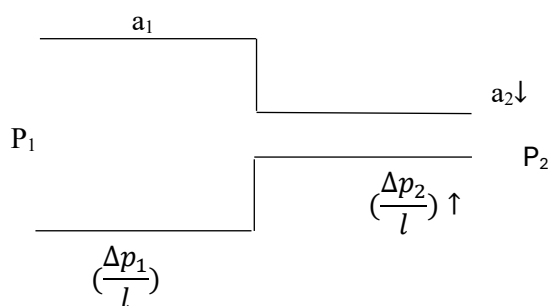
When tubes are connected in series, the volume flow rate (Q) passing through each tube is the same, similar to how current remains constant in a series electrical circuit.

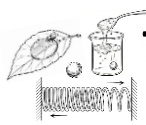


Since series,

$$\frac{\pi a_1^4 (P_1 - P_3)}{8 \eta l_1} = \frac{\pi a_2^4 (P_3 - P_2)}{8 \eta l_2}$$

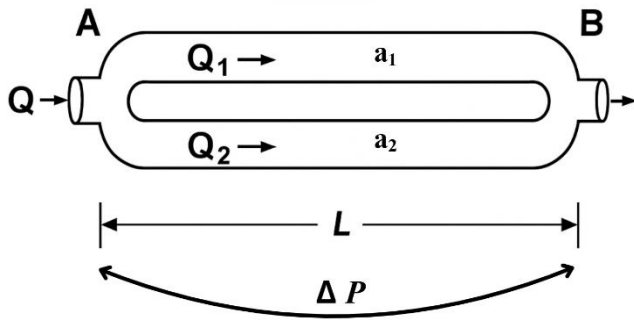
$$\frac{a_1^4}{l_1} (P_1 - P_2) = \frac{a_2^4}{l_2} (P_3 - P_2)$$





02) Tubes connected in parallel

When tubes are connected in parallel between two points, the pressure difference across each tube is the same.

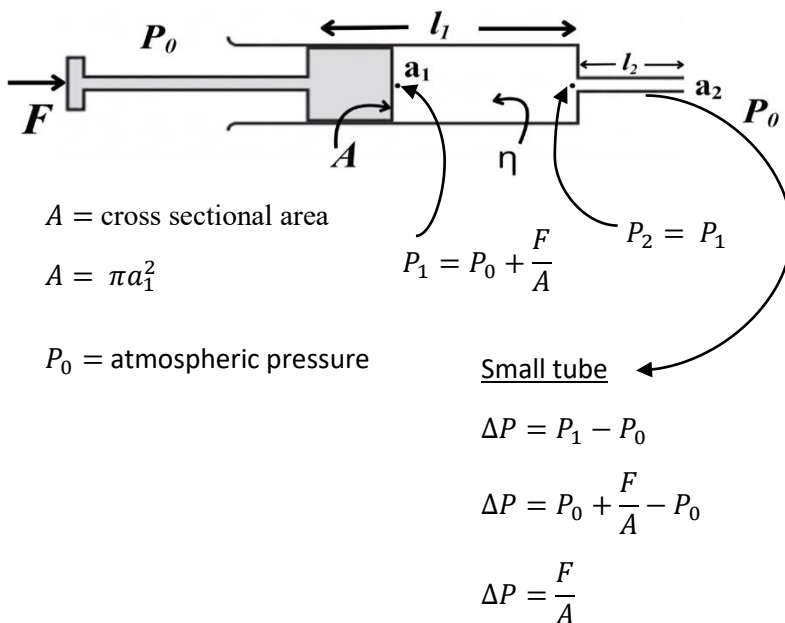


$$Q_1 = \frac{\pi a_1^4}{8 \eta} \left(\frac{\Delta P}{L} \right)$$

$$Q_2 = \frac{\pi a_2^4}{8 \eta} \left(\frac{\Delta P}{L} \right)$$

➤ Using a syringe to inject a liquid drug

01) Injecting a fluid into atmosphere



Poiseuille's equation for small tube

$$Q = \frac{\pi a_2^4}{8 \eta} \left(\frac{\Delta P}{l_2} \right)$$

$$Q = \frac{\pi}{8} \times \frac{a_2^4}{\eta} \times \left(\frac{F}{A l_2} \right)$$

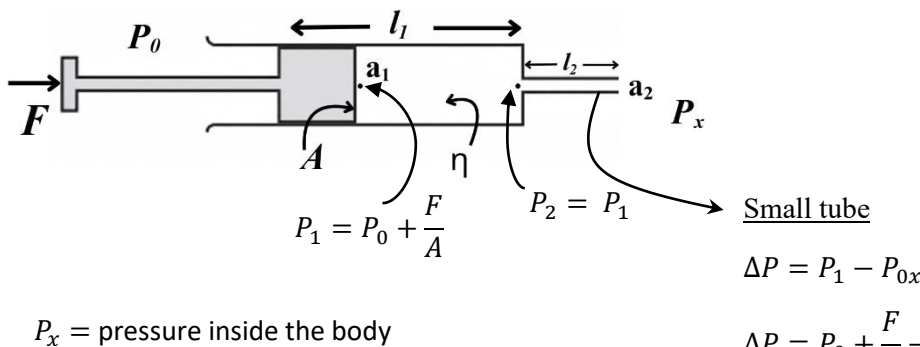
$$F = \frac{8 \eta A l_2 Q}{\pi a_2^4}$$

$$F = \frac{8 \eta \pi a_1^2 l_2 Q}{\pi a_2^4}$$

$$F = \frac{8 \eta a_1^2 l_2 Q}{a_2^4}$$

02) Releasing drugs into the body

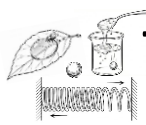
In such an instance, only difference is free end of injection pressure won't be equal to atmospheric pressure and it will equal to pressure inside the body (P_x)



Poiseuille's equation for a small tube

$$Q = \frac{\pi a_2^4}{8 \eta} \left(\frac{\Delta P}{l_2} \right)$$

$$Q = \frac{\pi}{8} \times \frac{a_2^4}{\eta} \times \left(P_0 + \frac{F}{A} - P_x \right)$$



Viscosity

Short Essay Questions

• **Beginner Level**

- (27) A liquid with viscosity η is placed between two horizontal plates. The bottom plate is fixed, and the top plate of area A moves at a constant velocity v . The distance between the plates is d . If a horizontal force F is applied to the top plate to maintain its motion, derive an expression for F in terms of η, v, A, d . A

Answer :- $\vec{F} = A\eta\left(\frac{v}{d}\right)$

- (28) A horizontal plate with an area of 10 cm^2 is placed on a table, and a 1 mm thick layer of glycerin is placed between the plate and the table. If the viscosity of glycerin is $2 \text{ kg m}^{-1}\text{s}^{-1}$, what horizontal force is required to pull the plate at a constant velocity of 10 m s^{-1} ?

Answer :- 20 N

- (29) Water with a density of 1000 kg m^{-3} flows through a horizontal tube. The cross-sectional area of one part of the tube is 10 cm^2 , and another part is 5 cm^2 . The pressure difference between these two parts is 375 Pa . Calculate the volume of water flowing through the tube per unit time.

Answer :- $5 \times 10^{-4} \text{ m}^3$

- (30) The terminal velocity of a spherical ball of radius r falling through a liquid with viscosity η is v . Find the terminal velocity of the same ball when it falls through a liquid with viscosity 3η .

Answer :- $V/12$

- (31) A plate with an area of 200 cm^2 is placed on an oil layer of 1 mm thickness. To move the plate at a constant velocity of 10 cm s^{-1} , a force must be applied. If the viscosity of the oil is $1.1 \times 10^{-3} \text{ N s m}^{-2}$, calculate the magnitude of the force.

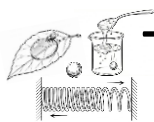
Answer : $2.2 \times 10^{-3} \text{ N}$

• **Average Level**

- (32) Blood flows through a horizontal capillary tube of radius 2 mm and length 20 cm . The volume flow rate is $2.5 \text{ cm}^3 \text{ s}^{-1}$. If the average viscosity of mercury is $4 \times 10^{-3} \text{ N s m}^{-2}$, calculate the pressure difference between the two ends of the tube.

Answer : $2.18 \times 10^2 \text{ N m}^{-2}$





- (33) Eight small spherical water droplets falling at terminal velocity combine to form a single large spherical drop. If buoyancy and upthrust are neglected, find the ratio of the terminal velocity of the large drop to the terminal velocity of a small drop.

Answer : $2^{2/3}$

- (34) Liquid with a viscosity of 0.8 N s m^{-2} is filled between two parallel horizontal plates 6 cm apart. A plate of area 0.01 m^2 is placed between these two plates and moved at a constant velocity of 1 m s^{-1} parallel to the plates. The distance from the top plate to this moving plate is 4 cm. Calculate the total viscous force acting on the plate.

Answer : 0.6 N

- (35) Two water droplets of equal radius fall through the air with a terminal velocity of 10 cm s^{-1} . After the two droplets merge into one and continue to fall, calculate the new terminal velocity.

Answer : 15.8 cm s^{-1}

- (36) A small sphere of radius r is released from height h above the surface of a viscous liquid. If the velocity of the sphere remains constant (reaches terminal velocity) immediately upon entering the liquid, show that h is proportional to r^4 . What is the relationship?

Answer : r^4

• **Master Level**

- (37) A spherical lead ball is dropped into a tall cylinder containing oil of height 20 cm and density 1800 kg m^{-3} . If it takes 30 s for the ball to travel that distance at terminal velocity, find the radius of the smallest ball. (Densities of water and coefficient of viscosity are 1000 kg m^{-3} and $0.0016 \text{ kg m}^{-1}\text{s}^{-1}$ respectively)

Answer : 10^{-2} mm

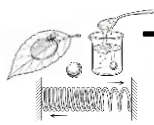
- (38) Two lead balls of radii 0.5 mm and 1.5 mm are released simultaneously from the top of a 1 m tall tube filled with oil. Find the time interval between them reaching the bottom. (Density of lead is 11300 kg m^{-3} , density of oil is 1200 kg m^{-3} , and viscosity of oil is $14.7 \text{ g cm}^{-1}\text{s}^{-1}$)

Answer : 3.9 minutes

- (39) A beaker is filled with a mixture of two liquids with densities 2.5 g cm^{-3} and 1.5 g cm^{-3} to a height of 8 cm. The viscosity of the mixture is $1.1 \times 10^{-3} \text{ N s m}^{-2}$. If a small spherical ball is dropped and takes 6 s to travel through the mixture, calculate the radius of the ball.

Answer : $1.1 \times 10^{-4} \text{ cm}$





- (40) 27 identical small spherical water droplets fall through the air with a terminal velocity of 0.15 m s^{-1} . If these droplets merge to form one large drop, calculate its terminal velocity.

Answer : 1.35 m s^{-1}

- (41) A charged oil droplet moves at a terminal velocity of $3 \times 10^{-4} \text{ m s}^{-1}$ in air. When a downward electric field of $5 \times 10^3 \text{ V m}^{-1}$ is applied, the droplet moves upward at the same terminal velocity. Calculate the radius and the charge of the droplet. (Density of oil and air are 900 kg m^{-3} and 1.30 kg m^{-3} ; viscosity of air is $1.83 \times 10^{-5} \text{ N s m}^{-2}$).

Answer : $1.66 \times 10^{-4} \text{ m}$, $3.442 \times 10^{-13} \text{ C}$

MCQ

• Beginner Level

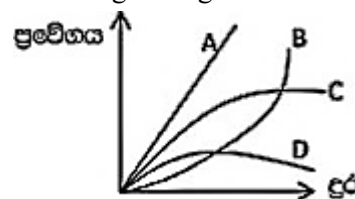
- (42) Consider the following statements regarding the flow of a liquid through a narrow tube.

- (A) Terminal velocity depends on the radius of the tube.
(B) The velocity of the liquid layer in contact with the tube wall is zero.
(C) The speed of the liquid does not depend on viscosity.

- (1) A only correct (2) B only correct (3) A and B correct
(4) A, B and C all correct (5) A, B and C all incorrect

- (43) The graph shows the variation of velocity (v) with distance (x) for a small sphere moving through a viscous medium. Which curve best represents this motion?

- (1) A (2) B (3) C (4) D (5) None of these



- (44) Under which of the following conditions is terminal velocity reached most slowly?

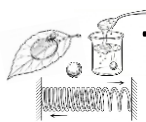
- (1) Small radius, high viscosity (η), and high density (ρ).
(2) Small radius, low density (ρ), and high viscosity (η).
(3) Large radius, low density (ρ), and high viscosity (η).
(4) Large radius, high density (ρ), and low viscosity (η).
(5) Small radius, high density (ρ), and high viscosity (η).

- (45) Consider a spherical object falling through a viscous medium. Which of the following is true?

- (A) The acceleration decreases until the object reaches terminal velocity.
(B) At terminal velocity, the weight of the object is balanced by the sum of upthrust and viscous force.
(C) Terminal velocity changes if the density of the medium changes.

- (1) A only (2) A and B only (3) A and C only (4) B and C only (5) A, B and C all





(46) Water flows streamline through two capillary tubes A and B of equal length and internal radii r and $2r$ respectively. The pressure difference across pipe A is P and the pressure difference across pipe B is P . What is the ratio of the volumetric flow rate of water flowing through A and B?

- (1) 1:4 (2) 1:8 (3) 1:16 (4) 1:32 (5) 1:64

(47) The volumetric flow rate of a viscous liquid flowing through a horizontal capillary tube as a continuous streamline flow is Q . Consider the following statements.

- A) Doubling the length of the pipe doubles Q .
B) Doubling both the length and diameter of the pipe increases Q by a factor of eight.
C) Doubling the pressure difference across the pipe doubles Q .

- (1) A only (2) B only (3) A and C only
(4) B and C only (5) A, B and C all

(48) A liquid flows through two unconnected capillary tubes under the same pressure difference. The ratio of their lengths is 1:2 and the ratio of their diameters is 2:1. What is the ratio of the flow rate of the liquid as it flows?

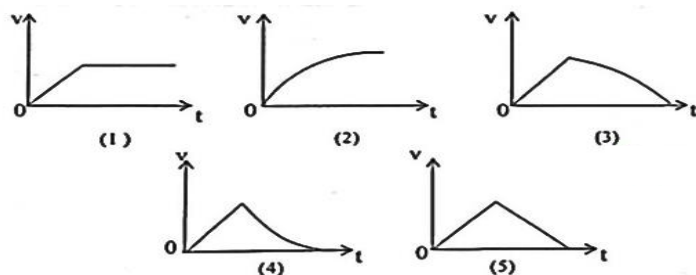
- (1) 2:1 (2) 4:1 (3) 8:1 (4) 16:1 (5) 32:1

• Average Level

(49) A spherical drop of oil rises from the bottom of a deep lake filled with water. After a while, it reaches its terminal velocity V_0 . Then it breaks into two identical spherical drops. If those two smaller drops also rise, what will be their terminal velocity?

- (1) $2^{(3/1)}V_0$ (2) $2^{(-2/3)}V_0$ (3) $2^{(3/2)}V_0$ (4) $2^{(1/3)}V_0$ (5) V_0

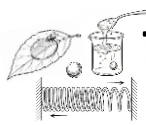
(50) A small sphere is dropped from rest from a height h onto the surface of a tall vessel of water. The density of the sphere is less than the density of water. The terminal velocity of the sphere to keep it freely in water is v_0 and $v_0 = \sqrt{2gh}$. What is the graph that represents the change in velocity v with time t as the sphere moves downward?



(51) If an air bubble moving upward in a reservoir with a terminal velocity x_0 breaks into three identical bubbles, what is the terminal velocity of one air bubble?

- (1) $3x_0$ (2) $\frac{x_0}{3}$ (3) $\frac{3^{\frac{2}{3}}}{x_0}$ (4) $3^{\frac{1}{3}}x_0$ (5) $3^{\frac{-2}{3}}x_0$





(52) The flow rate of a streamline flow in a capillary tube with radius a and a pressure difference P at the two ends is Q . If the radius is reduced to $a/2$ and the pressure difference is doubled, the new flow rate in the capillary tube is,

(53) A viscous fluid flows through a horizontal pipe of length l and diameter d at a pressure difference of ΔP , at a rate

- | | | | | |
|----------|---------------------|-------------------|--------------------|--------------------|
| (1) $4Q$ | (2) Q | (3) $Q/4$ | (4) $Q/8$ | (5) $Q/16$ |
| (1) V | (2) $\frac{V}{128}$ | (3) $\frac{V}{8}$ | (4) $\frac{V}{16}$ | (5) $\frac{V}{32}$ |

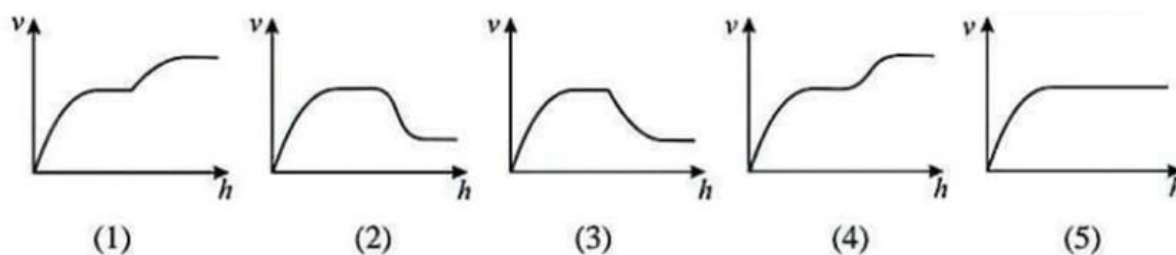
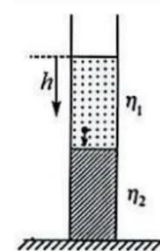
of V per second. If four pipes of length $2l$ and diameter $d/2$ are connected in series instead of the pipe, the pressure difference between the ends of the pipe remains the same ΔP . Now, what is the volume of fluid that flows through the pipe per second?

(54) A capillary tube of radius r and length l is connected in series with a capillary tube of radius $2r$ and length $2l$. A liquid flows through the tube. What is the ratio of the pressure difference across the first tube to the pressure difference across the second tube?

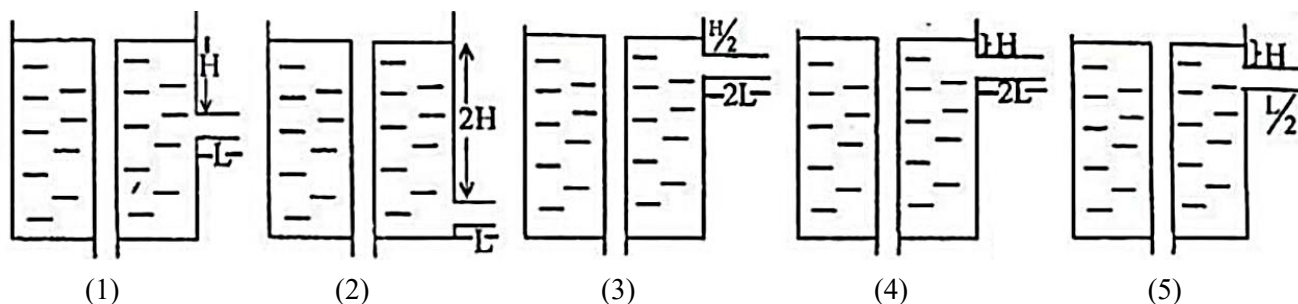
- | | | | | |
|-------|-------|-------|-----------|-----------|
| (1) 8 | (2) 2 | (3) 4 | (4) $1/8$ | (5) $1/4$ |
|-------|-------|-------|-----------|-----------|

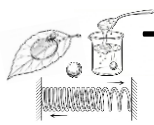
• Master Level

(55) A small sphere is released from the top of a tall vessel containing two liquids as shown in the figure. The viscosity of the liquid at the top of the vessel is η_1 and that of the liquid at the bottom is $\eta_2 (< \eta_1)$. Which of the following graphs best represents the change in its velocity v until it reaches the bottom? Assume that the depth of the liquids is sufficient for the sphere to attain its terminal velocity and that its density is greater than the density of both liquids.



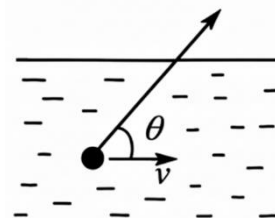
(56) When comparing the flow rate of water flowing through five pipes of the same cross-sectional area but different lengths and positions, which one will have the maximum?





(57) A small sphere of radius a and density σ is dragged by a string through a fluid of density ρ with a constant horizontal velocity. The inclination η of the string to the horizontal is θ . If the coefficient of viscosity of the fluid is η , what is the $\tan \theta$,

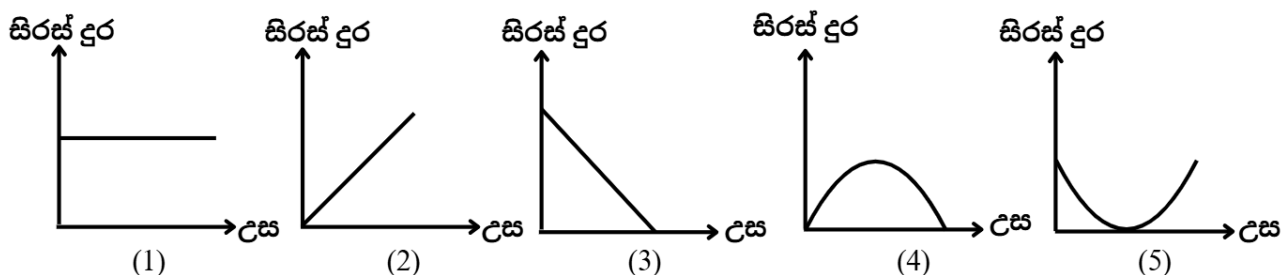
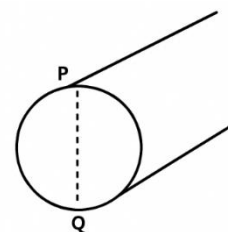
- (1) $\frac{2a^2 g(\sigma - \rho)}{9\eta}$ (2) $\frac{2a^2 \rho g}{9\eta v}$ (3) $\frac{2a^2 g(\sigma - \rho)}{3\eta}$
(4) $\frac{2a^2 \sigma g}{9\eta v}$ (5) $\frac{\sigma a \eta v}{(\sigma - \rho)}$



(58) Two spheres A and B are made of the same material; the volume of B is 8 times the volume of A. When they are allowed to fall freely in a viscous fluid, the terminal velocity of A is v_1 and the terminal velocity of B is v_2 . What is the terminal velocity of the system when the two spheres are connected to each other by a light string and placed in the same fluid?

- (1) $\frac{v_1 + v_2}{2}$ (2) $\frac{v_1 + v_2}{3}$ (3) $\frac{v_1 + v_2}{4}$ (4) $\frac{v_1 + 2v_2}{3}$ (5) $\frac{2v_1 + v_2}{3}$

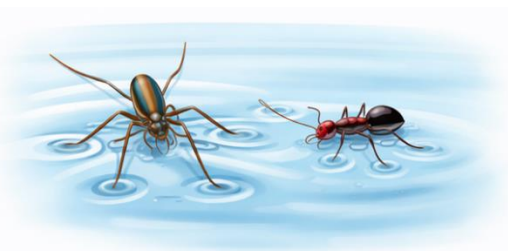
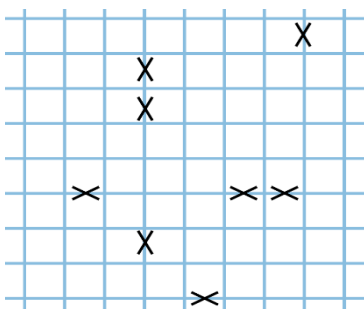
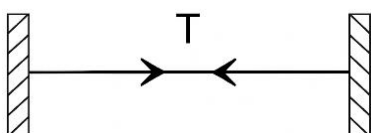
(59) As shown in the figure, a tube closed at one end has identical holes along a diameter, allowing a viscous fluid to flow through the entire tube and strike the closed surface. What correctly indicates how the horizontal distances to the point where the initial stream falls leaving each hole, from P to Q, vary with the height to the holes? (In the middle layer the viscous forces are minimum so the velocities are maximum),



Surface Tension

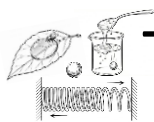
Just as tension builds up on a stretched string, tension is also stored within the surface when a surface is stretched. This is called surface tension.

Surface tension forces can be observed when a stretched liquid surface is considered and they are caused by intermolecular forces of attraction.



When identifying surface tension, two basic types of forces can be studied.

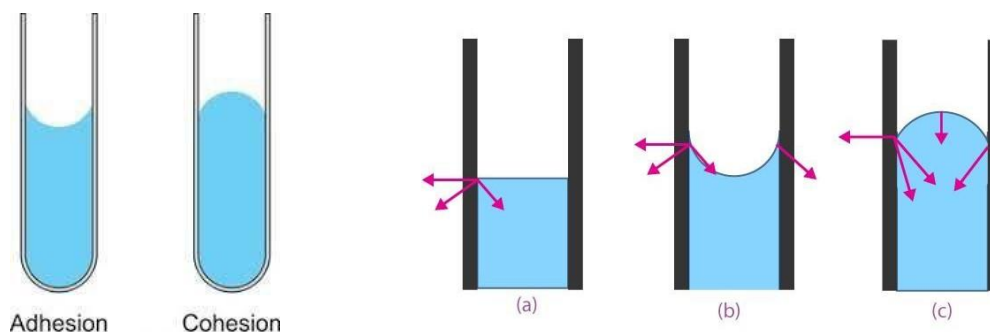




➤ Adhesive and cohesive forces

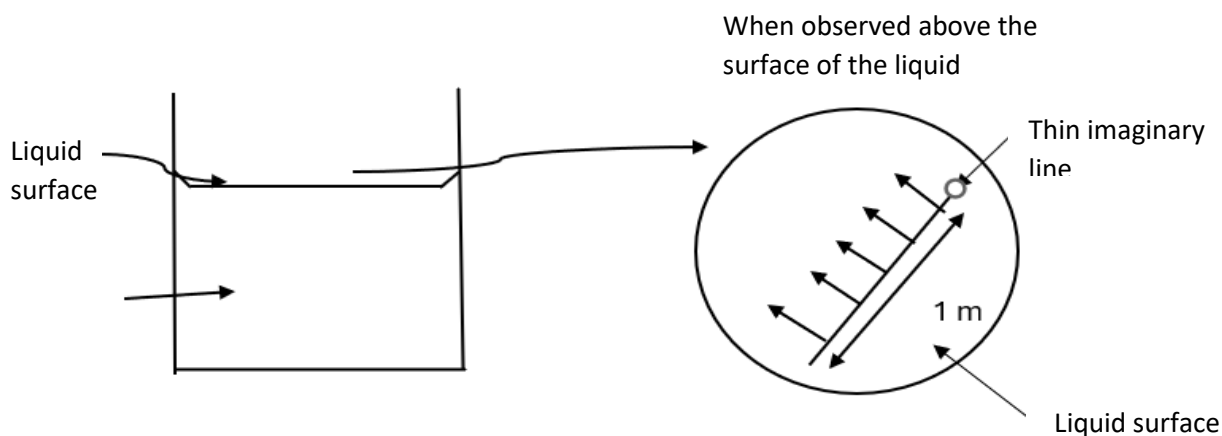
The attractive forces between dissimilar surfaces are called adhesive forces and the attractive forces between similar surfaces are called cohesive forces.

In contact between two objects, the nature of the collision between the two objects depends on which type of force is greater.

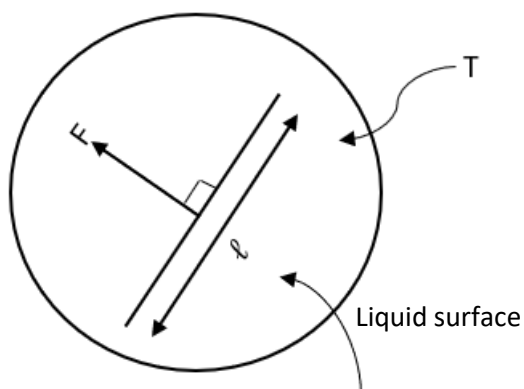


➤ Coefficient of surface tension (T or γ)

The force exerted by one side of the surface parallel to an imaginary line of unit length drawn on a surface, perpendicular to the surface is called the surface tension coefficient for the surface.



Relationship between surface tension force, coefficient of surface tension and the length on which the force is exerted



Surface tension force on a unit length = T

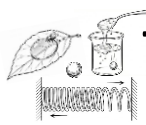
$\therefore$ surface tension force on a length $\ell = T\ell$

$$F = T\ell$$

$$T = \frac{F}{\ell}$$

Units of $T = \text{Nm}^{-1}$

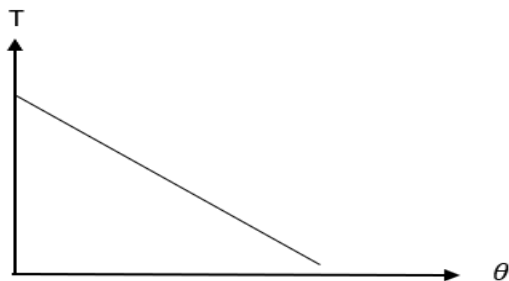
$$[T] = \frac{MLT^{-2}}{L} = MT^{-2}$$



Variation of the coefficient of surface tension with temperature

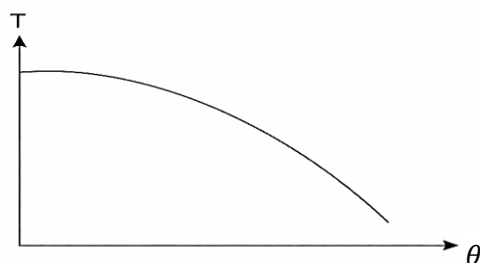
As the random motion of liquid molecules increases when the temperature of the liquid is increased, intermolecular forces between liquid molecules become weaker. Then the coefficient of surface tension of the liquid decreases.

1. Variation of the surface tension of water with temperature



As water is a polar molecule, the variation of surface tension of water with temperature is considerably higher when compared to other liquids.

2. Variation of surface tension of a liquid like oil with temperature



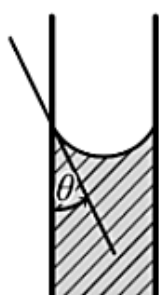
➤ Contact angle

When a liquid surface is touched with another surface, the angle measured in the liquid between the tangent drawn relative to the liquid surface and the colliding surface is the contact angle.

In general, the surface tension coefficient of a liquid is a value that depends on the liquid.

Adding soap to water $T \downarrow$

Adding salt to water $T \uparrow$



$$\theta < 90^\circ$$



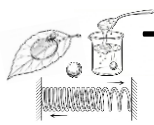
$$\theta > 90^\circ$$



$$\theta = 0$$

Surface tension forces from the liquid to the vessel or vessel to liquid always act along this tangent.

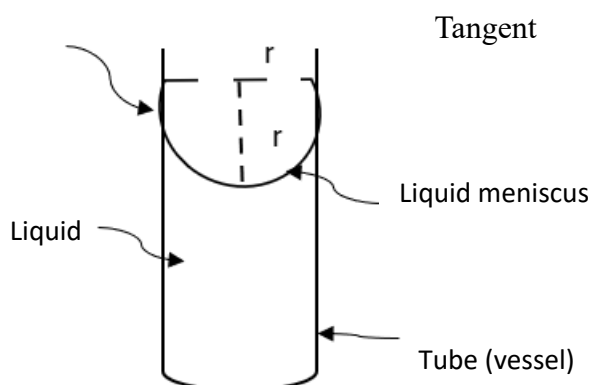




Liquids with zero surface tension

Such liquids are called liquids that wets the vessel.

Eg: Water and alcohol are liquids that wet glass.



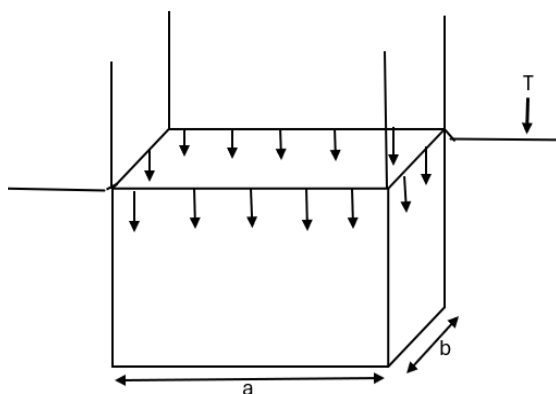
When a liquid with zero contact angle is inserted in a tube, the radius of the liquid meniscus formed is equal to the radius of the tube.

$$\text{Radius of the liquid meniscus} = \text{Radius of the tube}$$

Obtaining surface tension forces acting on objects with different shapes

Surface tension force acting along the edges when a part of an object with a rectangular cross section is inserted in a liquid is,

1. When the contact angle is zero

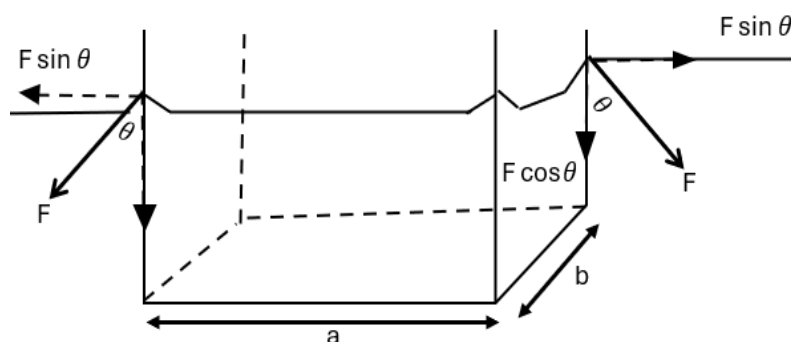


Total surface tension force,

$$F = T (2a + 2b)$$

$$F = 2T (a + b)$$

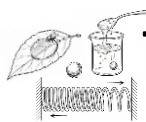
2. When the contact angle is an acute angle,



Total surface tension force,

$$F = T (2a + 2b) \cos \theta$$

$$F = 2T (a + b) \cos \theta$$



When a hollow glass tube of external radius a and internal radius b is inserted partly in a liquid

1. When the contact angle is zero

Total surface tension force,

$$\downarrow F = T (2\pi a + 2\pi b)$$

$$\downarrow F = 2\pi T (a + b)$$

2. When the contact angle is acute,

Total surface tension force,

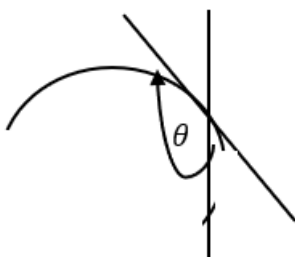
$$\downarrow F = 2\pi T (a + b) \cos \theta$$

3. When the contact angle is 90° ,

Total surface tension force,

$$\downarrow F = 0$$

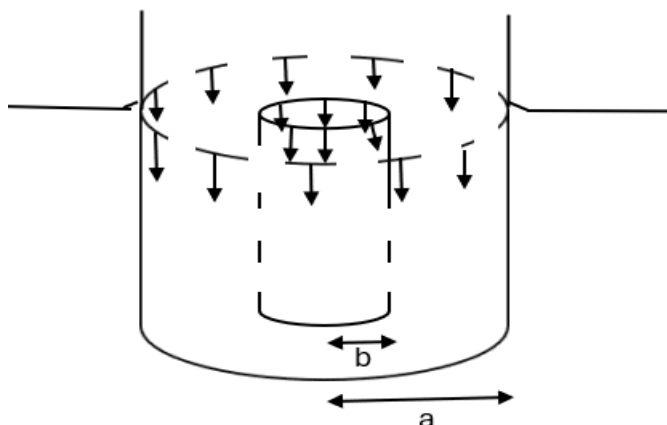
4. When the contact angle is obtuse,



Total surface tension force,

$$\uparrow F = 2\pi T (a + b) \cos \alpha$$

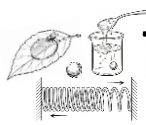
$$\uparrow F = 2\pi T (a + b) \cos(180 - \phi)$$



Examples:

- 01) A plate 2 mm thick, 6 cm long and 4 cm wide is placed with one of its largest faces in contact with a water surface. What is the downward force exerted on the plate by the water surface? The contact angle of water is zero and the surface tension is $7 \times 10^{-2} \text{ Nm}^{-1}$.
- 02) A clean square glass plate of side 8 cm is 2 mm thick. The mass of the plate is first weighed in air. Then a string is tied to one vertex of the plate and the plate is slowly immersed in water so that one diagonal of the plate is on the surface of the water. If the surface tension of water is $7.2 \times 10^{-2} \text{ Nm}^{-1}$, the density is 1000 kgm^{-3} and the contact angle between the water and the glass is zero, calculate the decrease in the apparent mass of the plate.

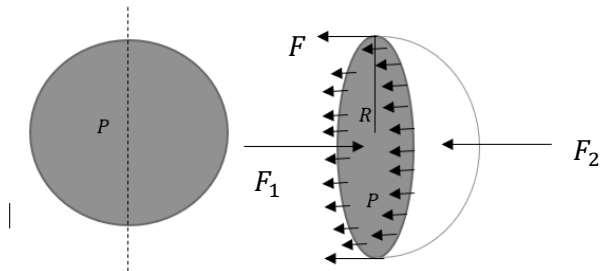




- 03) A microscope slide 10 cm long and 1 mm thick is suspended from one arm of a chemical balance with its long side horizontal. Now the balance is balanced and a container containing alcohol is placed below the microscope slide and the container is carefully raised until the surface of the alcohol just touches the bottom surface of the slide. When the slide comes into contact with the alcohol, the arm containing the slide lowers and a mass of 0.63 g has to be added to the other arm to and balance it again. Calculate the surface tension of alcohol.

➤ Deriving an expression for the difference between internal and external pressure values of a soap bubble and a water drop

Consider a soap bubble of radius r .



F_1 = Force due to internal pressure

F_2 = Force due to external pressure

F = Surface forces

Forces act on every infinitesimal cross section within the bubble due to pressure. If the resultant force on the semi circular portion due to internal pressure is F_1 ,

$$\vec{F}_1 = P_1 \times \frac{1}{2} \pi r^2 \times 2$$

$$\vec{F}_1 = P_1 \times \pi r^2$$

Here the surface area on which pressure forces are created is not the total area of the hemisphere but the area of the circular portion of radius r .

If the force on the hemispherical portion due to external force is F_2 ,

$$F_2 = P_2 \pi r^2$$

If surface tension force along the outer edge is F ,

$$F = T \times 2\pi r \times 2 \text{ (As a soap membrane has 2 surfaces)}$$

When horizontal force equilibrium of the hemisphere is considered,

$$F_1 = F + F_2 \quad P_1 - P_2 = \frac{4T}{r}$$

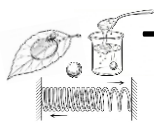
$$F_1 - F_2 = F \quad \Delta P = \frac{4T}{r}$$

$$P_1 \pi r^2 - P_2 \pi r^2 = T \times 2\pi r \times 2$$

$$(P_1 - P_2)r = 4T$$

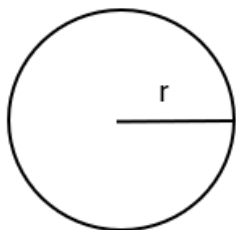
As the radius of the bubble is increased, excess pressure decreases.

$$\text{When } r \rightarrow \infty \quad \Delta P \rightarrow 0$$

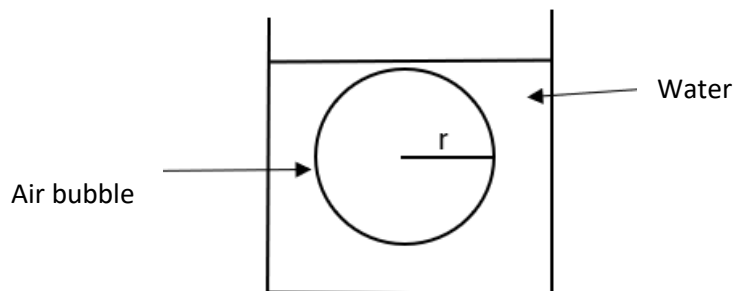


As a soap bubble has air on both internal and external sides, 2 surfaces subjected to tension can be identified.

When the excess pressure for a water drop or an air bubble in water is considered, it is taken as follows as they don't have two surfaces.

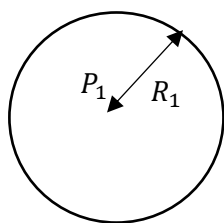


Water drop

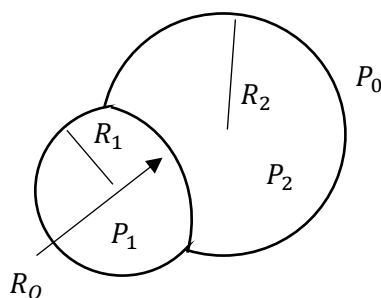


$$\Delta P = \frac{2T}{r}$$

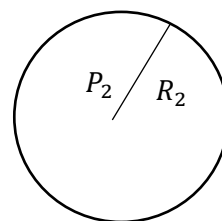
➤ Finding the radius of the common meniscus when two soap bubbles are added



$$P_1 - P_0 = \frac{4T}{R_1}$$



$$P_1 - P_2 = \frac{4T}{R_0}$$



$$P_2 - P_0 = \frac{4T}{R_2}$$

If the radius of the new meniscus is R_0 ,

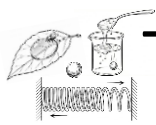
$$P_1 - P_0 - P_2 + P_0 = \frac{4T}{R_1} - \frac{4T}{R_2}$$

$$P_1 - P_2 = \frac{4T}{R_1} - \frac{4T}{R_2}$$

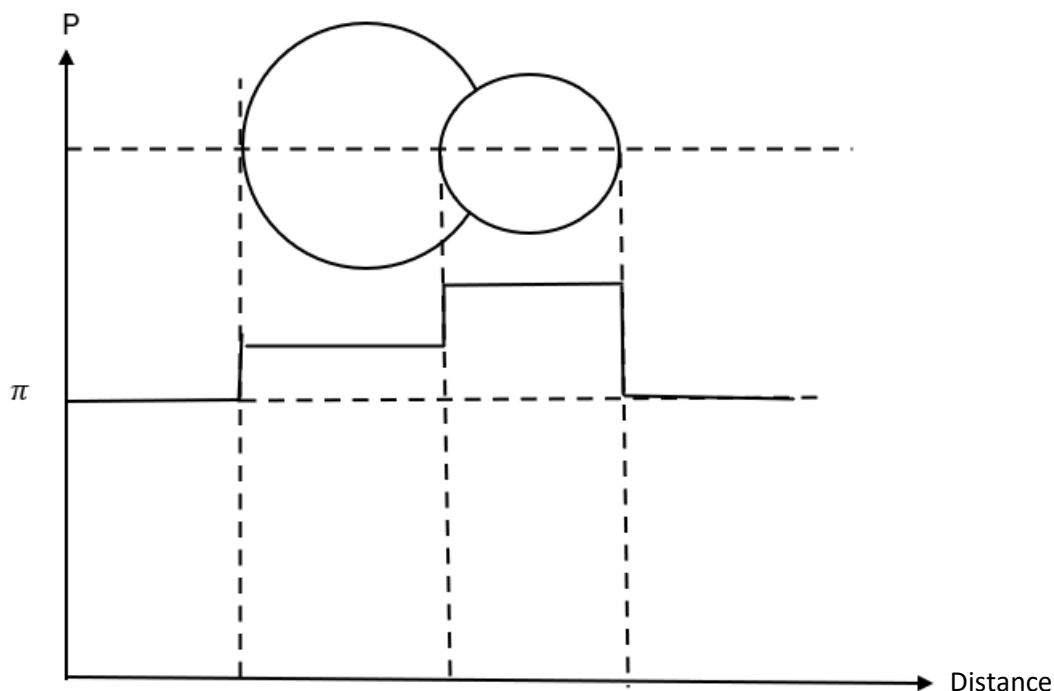
$$\frac{4T}{R_0} = \frac{4T}{R_1} - \frac{4T}{R_2}$$

$$\frac{1}{R_0} = \frac{1}{R_1} - \frac{1}{R_2}$$





Variation of the pressure with distance when two liquid (soap) bubbles are connected

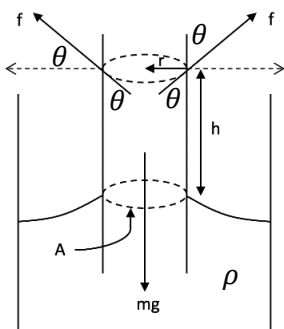


Examples:

- 1) Calculate the excess pressure difference between a water droplet and a soap bubble of radius 5 mm. The surface tension values of water and soap solution are $7 \times 10^{-2} \text{ Nm}^{-1}$ and $2.5 \times 10^{-2} \text{ Nm}^{-1}$ respectively.
- 02) A soap bubble of radius 2 mm is inflated inside another soap bubble of radius 3 mm. Calculate the radius of a soap bubble whose pressure is equal to the pressure inside the smaller soap bubble.

➤ Capillary rise

It has been observed that when a thin vertical tube is immersed in a liquid container, the water level in the tube rises. In this case, the weight of the water column is supported by the resultant vertical surface tension force exerted by the tube on the water column.



f = Force by the glass tube

m = cylindrical water volume

$$= (A \times h) \rho$$

$$\uparrow F_T = T \times 2\pi r \times \cos \theta$$

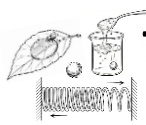
$$F_T = mg$$

$$T \times 2\pi r \times \cos \theta = (A \times h) \rho g$$

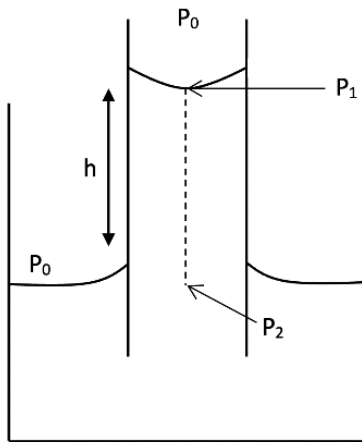
$$T \div 2\pi r \cos \theta = \pi r^2 \times h \rho g$$

$$\uparrow h = \frac{2T \cos \theta}{r \rho g}$$





- Obtaining an expression for the capillary rise considering the pressure difference on both sides of the meniscus



$$P_0 - P_1 = \frac{4T}{R}$$

By hydrostatics,

$$P_2 = P_1 + h\rho g$$

Same horizontal level

$$P_0 = P_1$$

$$P_0 = P_1 + h\rho g$$

$$P_0 - P_1 = h\rho g$$

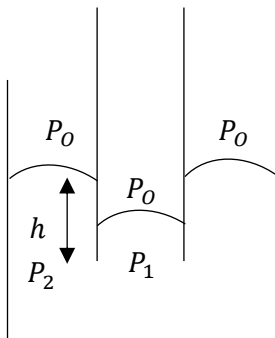
$$\frac{2T}{R} = h\rho g$$

$$\frac{2T}{R\rho g} = h$$

$$\frac{2T \cos \theta}{r\rho g} = h$$

- Deriving an expression for the capillary depression when a capillary tube is immersed in a liquid

When a liquid and a tube are in contact, if the cohesive forces are stronger inside the liquid, capillary depression can be observed.



Surface tension

$$P_1 - P_0 = \frac{2T}{R}$$

Hydrostatics

$$P_0 + h\rho g = P_2$$

$$P_2 = P_1$$

$$P_0 + h\rho g = P_1$$

$$h\rho g = P_1 - P_0$$

$$h\rho g = \frac{2T}{R}$$

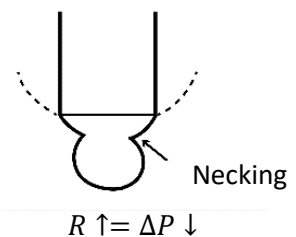
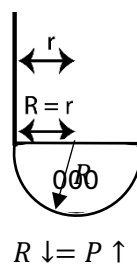
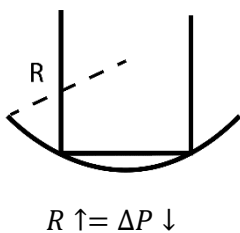
$$h = \frac{2T}{R\rho g}$$

$$h = \frac{2T \times (-\cos \theta)}{r\rho g}$$

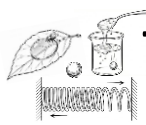
$$h = \frac{-\cos \theta 2T}{r\rho g}$$

- Finding the maximum height of a body of column that can be held in a container with a hole

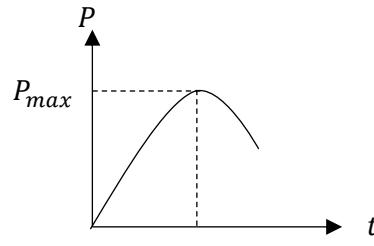
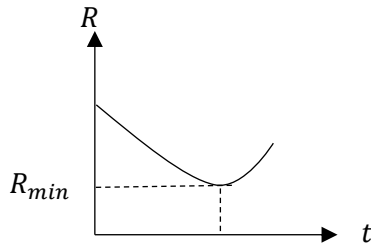
Let's consider a case where a drop of water drips from the edge of a faucet.



$$\Delta P = \frac{2T}{R}$$



The pressure difference across the meniscus is minimum when the radius is maximum.



- The minimum value that can be obtained for the radius of the meniscus is the radius of the tube. Then there is a maximum pressure difference.
- Thus, the following expressions can be obtained for the maximum mass of water that can be retained.

Maximum volume of water at equilibrium = (V) Water mass = (m)

$$= \frac{4}{3}\pi R^3 \times \frac{1}{2}$$

$$= \frac{4}{3}\pi r^3 \times \frac{1}{2}$$

$$= v \times \rho_w$$

$$= \frac{4}{3}\pi R^3 \times \frac{1}{2} \times \rho_w$$

$$= \frac{2}{3}\pi R^3 \rho_w$$

$$r = R$$

R = radius of the meniscus

r = radius of the tube



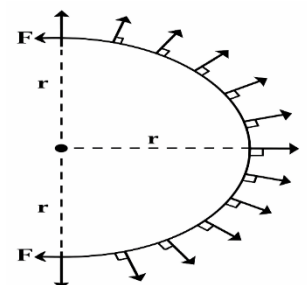
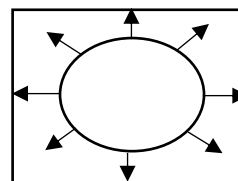
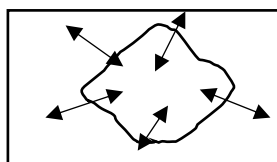
Examples:

- 01) A glass U-tube has two arms of radii 2 cm and 1 cm, and contains a liquid of density 1270kgm⁻³ and surface tension 4.5x10⁻²Nm⁻¹. Calculate the difference in liquid levels in the two arms, assuming that the contact angle between the liquid and the glass is zero.
- 02) When a capillary tube is immersed in water, water rises to a height of 12 cm in the tube. When it is immersed in alcohol, alcohol rises to a height of 5.5 cm in the tube. If the densities of water and alcohol are 1000kgm⁻³ and 800kgm⁻³ respectively, calculate the surface tension of water. Assume that the contact angle between alcohol and glass and between water and glass is zero.

➤ Calculating the tension that a rubber band experiences when the inner surface of the band is broken by placing it on a soapy surface

When a rubber band is placed on a soap film, the rubber band is initially not under tension because the film is present on both the inside and outside. But when the inner surface is broken, the rubber band is under tension because only the outer film remains to exert the surface tension force.

By getting the horizontal forces when the equilibrium of half of the rubber band is



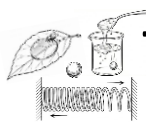
Sum of the tension forces acting towards = Resultant surface tension force on the semicircular portion

(As the soap membrane has two surfaces)

$$2F = 2 \times T \times 2r$$

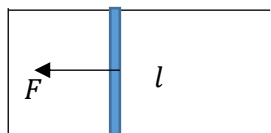
$$F = 2Tr$$

Here the surface tension force on the semicircular portion is equal to the surface tension force that acts on a length equal to the diameter of the loop.



➤ Surface energy

Using a soap film spread over a wire frame, the following expressions can be constructed for surface tension:.



$$\text{Work done to pull.}(W) = F_o \cdot S$$

$$= F_o \cdot x$$

$$= 2Tl \times x$$

Work is stored as surface energy.

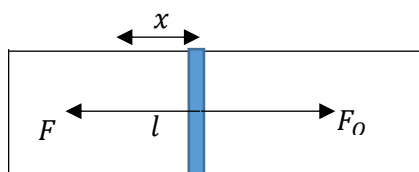
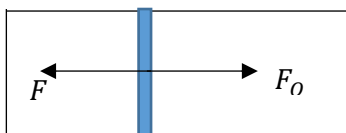
$$\Delta E = W$$

$$\Delta E = 2Tl \times 2$$

$$\Delta E = T \times 2lx$$

$$\Delta E = T \times \Delta A$$

$$\text{surface energy; } E = T \times A$$



$$\text{Free surface energy} = \frac{\text{surface energy}}{\text{Area}}$$

Water drop

$$E = T \times 4\pi r^2$$

Soap bubble

$$E = T \times (4\pi r^2 + 4\pi r^2)$$

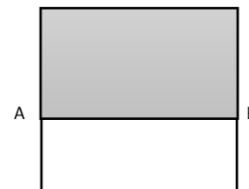
$$= T \times 8\pi r^2$$

$$\frac{E}{A} = T \times \frac{A}{A}$$

Free surface energy

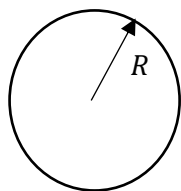
Coefficient of surface

(Q) A soap film is made in a frame with a movable rod AB as shown in the figure. What should be the cross-sectional diameter of rod AB to maintain equilibrium? If the work done by rod AB in moving 1 cm down the frame is $4.5 \times 10^{-5} \text{ J}$, calculate the length of the rod. The density of the material of which the rod is made is 8600 kgm^{-3} and the surface tension of the soap is $4.5 \times 10^{-2} \text{ Nm}^{-1}$.

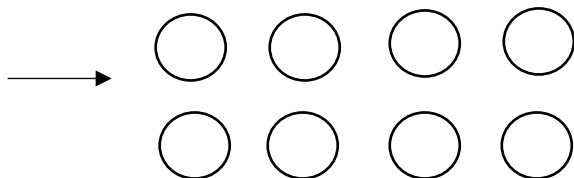


➤ The work done in splitting a water droplet into several water

Find the work done in breaking a water drop of radius R into 8 identical drops. Let the surface tension be T.



$$E_1 = T \times 4\pi R^2$$



$$E_2 = 8 \times 4\pi r^2 T$$

By volume,

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r^3 \times 8$$

$$R^3 = 8r^3$$

$$R = 2r$$

$$W = E_1 - E_2$$

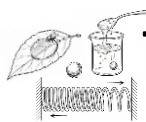
$$= 8 \times 4\pi r^2 T - 4\pi R^2 T$$

$$= 8 \times 4\pi \times (r^2) T - 4\pi R^2 T$$

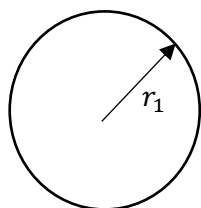
$$= 8\pi R^2 T - 4\pi R^2 T$$

$$W = 4\pi R^2 T$$

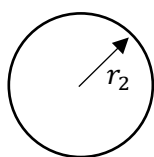
$$r = \frac{R}{2}$$



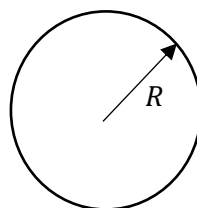
➤ Two water drops brought together under isothermal conditions to form a single bubble.



Soap



Soap



Principle of energy conservation,

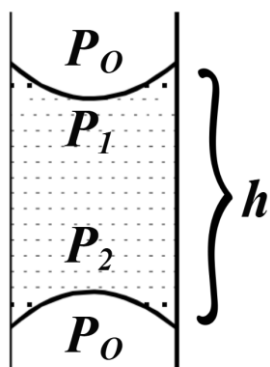
$$T \times 8\pi r_1^2 + T \times 8\pi r_2^2 = T \times 8\pi R^2$$

$$r_1^2 + r_2^2 = R^2$$

$$R = \sqrt{r_1^2 + r_2^2}$$

➤ Holding a column of liquid in a vertically placed capillary tube

In such a case, the following expressions can be obtained for the height of the liquid column.



$$P_0 - P_1 = \frac{2T}{R_1} \text{ ————— ①}$$

$$P_2 = P_1 + h\rho g$$

$$P_2 - P_1 = h\rho g \text{ ————— ②}$$

$$P_0 - P_2 = \frac{2T}{R_2} \text{ ————— ③}$$

$$\text{①} - \text{②} \text{ ka;}$$

$$P_0 - P_2 = \frac{2T}{R_1} - h\rho g \text{ ————— ④}$$

$$\text{③} - \text{④} \text{ ka;}$$

$$\frac{2T}{R_2} = \frac{2T}{R_1} - h\rho g$$

$$h\rho g = \frac{2T}{R_1} - \frac{2T}{R_2}$$

$$h = \frac{2T}{\rho g} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Surface tension

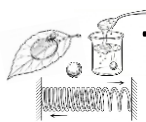
Short Essay Questions

• Beginner Level

(60) What is the excess pressure inside a bubble of radius r created by a soap solution of surface tension T ?

Answer :- $\frac{4T}{r}$



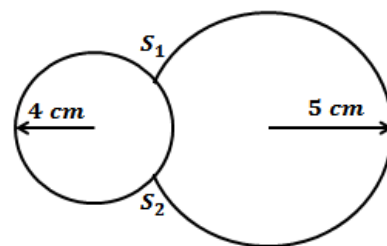


- (61) A thin film of soap is held in equilibrium by a force of 1.5×10^{-2} N in a U-shaped wire frame as shown in the figure. What is the surface tension of the soap solution?

Answer : $2.5 \times 10^{-2} \text{ N m}^{-1}$

- (62) Two soap bubbles of radii 4 cm and 5 cm are fused together as shown in the figure. Find the radius of the common surface of the two bubbles.

Answer : 20 cm



- (63) Calculate the amount of work done against surface tension forces in forming a soap bubble of radius 0.5 cm. Let the surface tension of the soap solution be $2.14 \times 10^{-2} \text{ N m}^{-1}$.

Answer : $6.725 \times 10^{-6} \text{ J}$

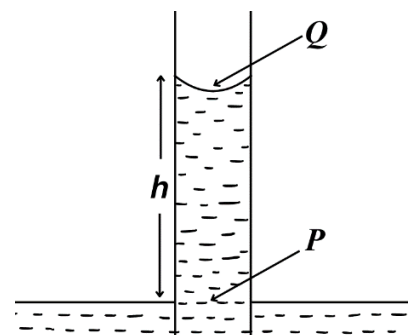
- (64) A plate 2 mm thick, 6 cm long and 4 cm wide is placed with one of its largest faces in contact with a water surface. What is the downward force exerted on the plate by the water surface? The contact angle of water is zero and the surface tension is $7 \times 10^{-2} \text{ N m}^{-1}$.

Answer : $8.68 \times 10^{-3} \text{ N}$

• Average Level

- (65) A capillary tube of radius r is immersed in a liquid of density d and surface tension T . The contact angle of the liquid is zero and the atmospheric pressure is π . Find the pressure difference between the two points P and Q.

Answer : $\Delta P = h d g$



- (66) Water rises to a height of 2.0 cm in a capillary tube. How high will water rise in a capillary tube with a radius $1/3$ the radius of the previous capillary tube?

Answer : 6 cm

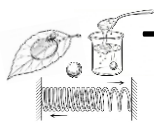
- (67) Calculate the amount of work required to break a water droplet of radius 0.5 cm into 125 drops under isothermal conditions. The surface tension of water is $7 \times 10^{-2} \text{ N m}^{-1}$.

Answer : $1.095 \times 10^{-2} \text{ J}$

- (68) The radii of the two arms of a U-tube are 1 mm and 3 mm. The arms of this tube are placed vertically and filled with water. If the contact angle of the water is zero, the surface tension is $7.2 \times 10^{-2} \text{ N m}^{-1}$ and the density is 1000 kg m^{-3} , find the difference between the liquid levels.

Answer : 14.4 mm





- (69) 1000 identical water droplets of radius 1 mm coalesce into a single drop. If its surface tension is 0.07 N m^{-1} , calculate the surface energy released in this process.

Answer : $7.92 \times 10^{-4} \text{ J}$

- (70) When a capillary tube of radius r is immersed in a liquid of surface tension T and density ρ , a column of liquid rises to a height h inside the tube. (Here the contact angle is θ). When the tube is taken out vertically and a liquid column of height h is inside it, the radius of the liquid meniscus at the lower end of the tube is,

Answer : $\frac{2Tr}{(2T \cos \theta - h\rho g r)}$

• Master Level

- (71) A drop of water of volume V is placed between two rectangular glass plates and the plates are pressed together so that it spreads over the entire area A of the plate. If the contact angle is zero, and the surface tension is T , what is the minimum perpendicular force that must be applied to one of the plates to separate the plates from each other??

Answer : $\frac{2TA^2}{V}$

- (72) A cylindrical sieve has n number of holes of radius r in its horizontal bottom. The area of the bottom is A . Find the maximum mass of water of density ρ and surface tension T that can be carried without leaking through the holes is,

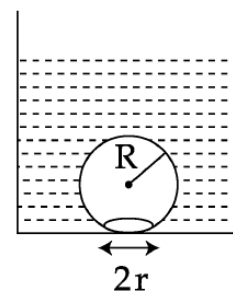
Answer : $\frac{2\pi r^3 \rho n}{3}$

- (73) A three-dimensional rubber ring has an unstretched radius of 3.8 cm. When this is placed in a soap film and the film inside the ring is broken, the new radius of the ring is 3.9 cm. When the ring is torn at one point, a force of 0.28 g increases the natural length of the ring by 1 cm. Calculate the surface tension of the soap solution.

Answer : $2.25 \times 10^{-2} \text{ N m}^{-1}$

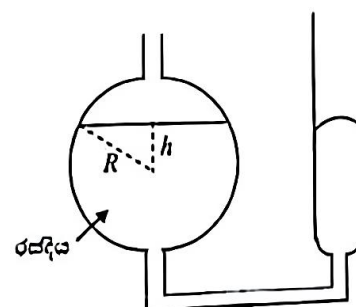
- (74) Air bubbles formed at the bottom of a hot water container separate from the bottom and rise. As shown in the figure, before rising, the air bubbles are connected to the bottom of the container by a circular region of radius r . The air bubble is spherical in shape with radius R . $R \gg r$ Find r at the moment when the air bubble separates from the bottom. (The coefficient of surface tension of water is T and the density is ρ .)

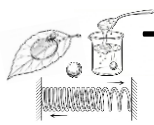
Answer : $r = R^2 \sqrt{\frac{2\rho g}{3T}}$



- (75) The setup shown in the figure is used to measure the angle of contact between mercury and glass. The mercury contained in a spherical glass bulb of radius of curvature R is exactly horizontal at a height h from the center at the given moment. Find the angle of contact between mercury and glass using the given symbols.

Answer : $\theta = \cos^{-1} \left(\frac{h^2 - R^2}{R^2} \right)$



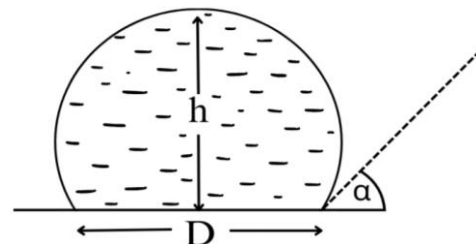


- (76) As shown in the figure, water is poured into a vessel to a height of 2 mm and a piston with a cross-sectional area of 4 cm^2 and a mass of 5 g is placed above it in contact with the water. The container has a hole with a radius of 0.5 mm at the bottom. What is the maximum mass that can be placed on the piston so that no water escapes through the hole? (Take the surface tension of water as $7 \times 10^{-2} \text{ N m}^{-1}$)

Answer : 5.4 g

- (77) If the density of this liquid drop is ρ and the surface tension is T , then express its weight (W) in terms of T , h , ρ , g , D and α

Answer : $W = \rho g h - T \pi D \sin \alpha$



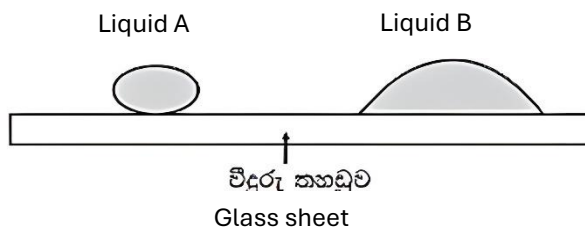
MCQ

• Beginner Level

- (78) The amount of work done to increase the area of a soap bubble from 50 cm^2 to 100 cm^2 is $2.4 \times 10^{-4} \text{ J}$. Surface tension of the soap solution is ,
- (1) $2.4 \times 10^{-2} \text{ N m}^{-1}$ (2) $4.8 \times 10^{-2} \text{ N m}^{-1}$ (3) $2.4 \times 10^{-4} \text{ N m}^{-1}$
(4) $4.8 \times 10^{-4} \text{ N m}^{-1}$ (5) $2.4 \times 10^{-6} \text{ N m}^{-1}$

- (79) The figure shows the shape of two drops of liquid A and B on a glass sheet. Consider the following statements.

- a) The density of that liquid A is greater than the density of the liquid B.
b) The surface tension of liquid A is greater than the surface tension of liquid B.
c) The contact angle of liquid A with glass is greater than the contact angle of liquid B.



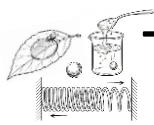
Which of the above statement/s is/are always true?

- (1) a only (2) b only (3) a and b only
(4) b and c only (5) All a, b and c
- (80) A liquid in a container does not wet the walls of the container. Its contact angle is ,
- (1) Lesser than 90° (2) 90° (3) 0°
(4) Greater than 90° (5) 180°
(6)
- (81) Consider the following statements.
- a) The surface tension of a liquid increases with temperature.
b) Liquids tend to minimize their surface area as much as possible.
c) The potential energy at the surface molecules of a liquid is greater than the potential energy of molecules inside the liquid

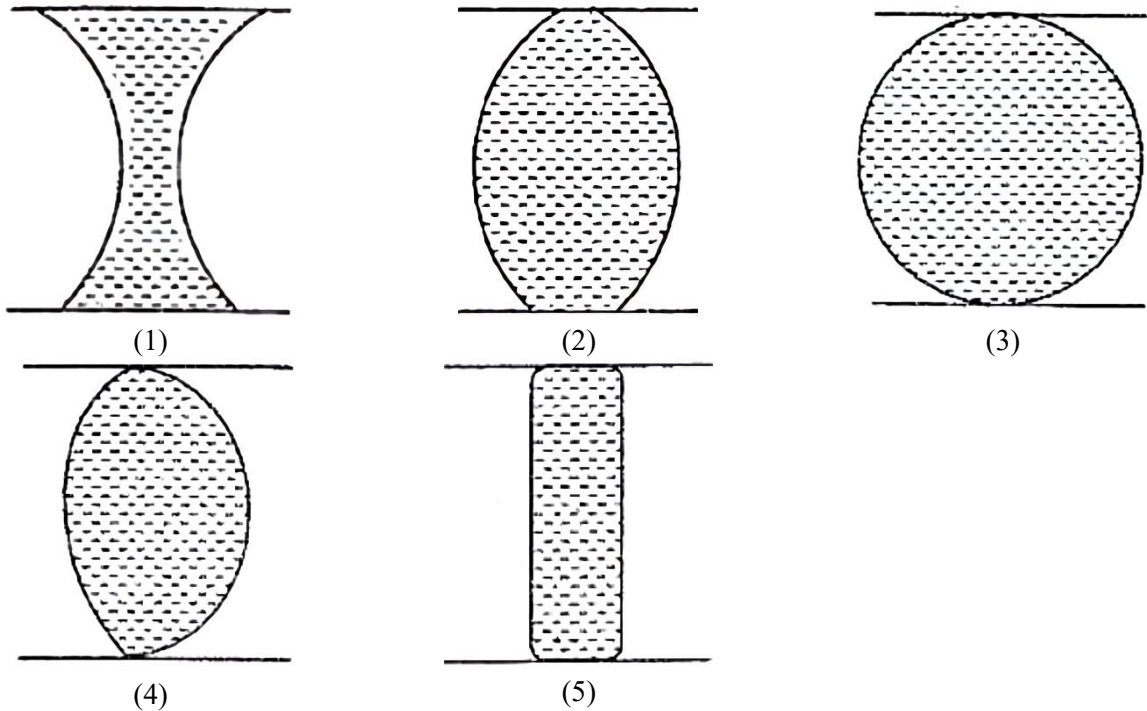
The correct statement/s from the above is/are

- (1) a only. (2) b only. (3) c only.
(4) a and b only (5) b and c only.





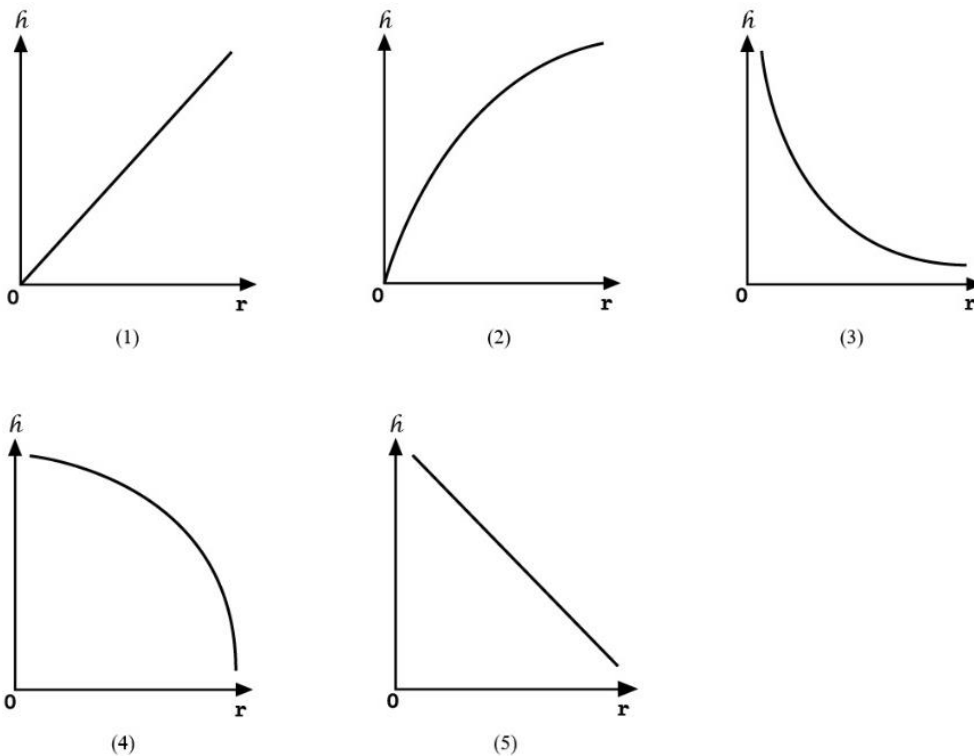
(82) The shape of a drop of water held between two sheets of glass is best represented by ,

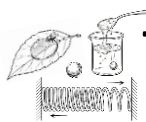


(83) The capillary rise of water in a glass tube with an internal radius of 1 mm is h . Find rise of water in a glass tube of radius 1 cm.

- (1) 0 (2) $h/10$ (3) $h/5$ (4) $h/2$ (5) h

(84) Which of the graphs below correctly shows the variation between capillary rise h and the inner radius r of the capillary tube,



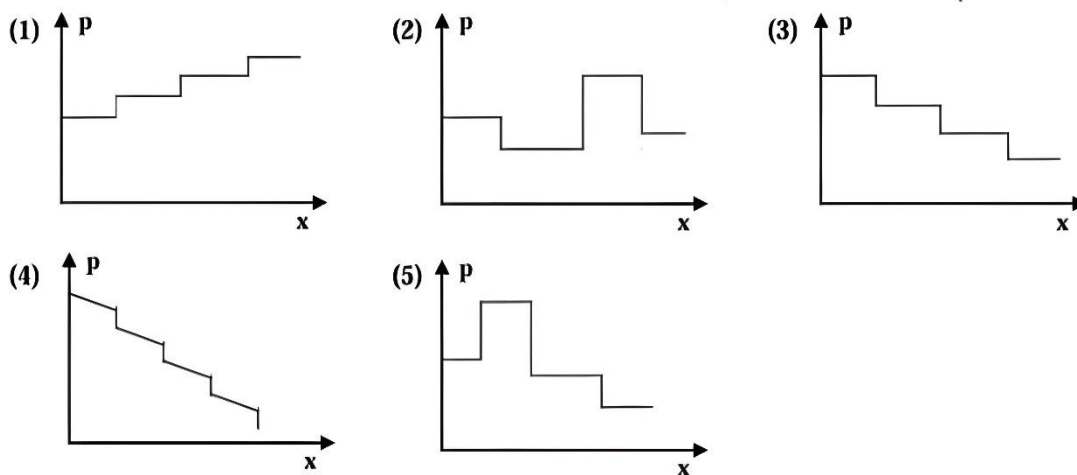
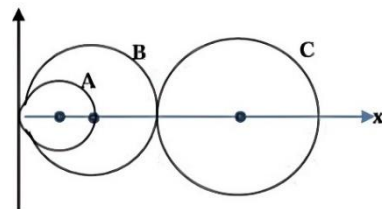


• **Average Level**

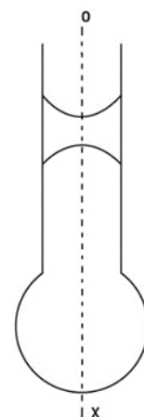
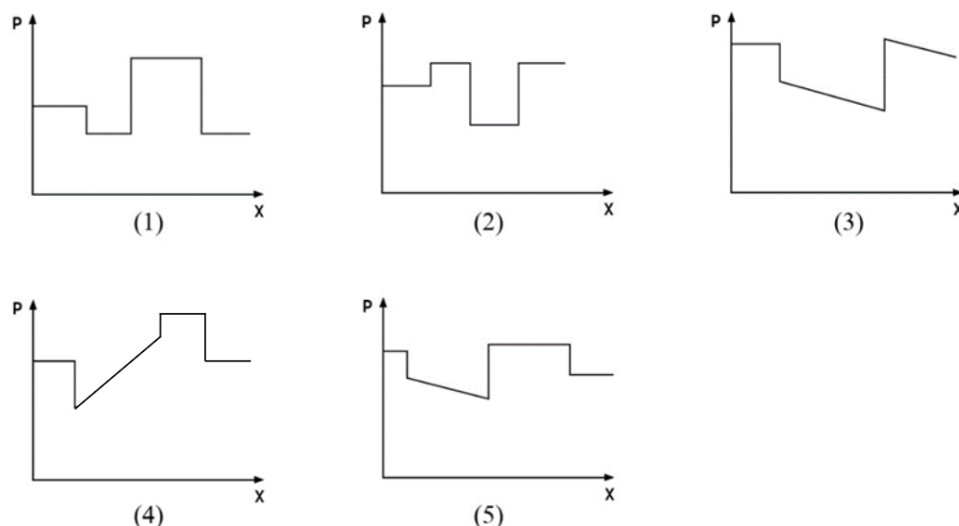
- (85) When a capillary tube was immersed in a container of water, capillary rise was x on the surface of the Earth. What is the new capillary rise when this vessel with the capillary tube is placed at the bottom of a crater of depth h formed by an earthquake? (R = radius of earth)

(1) $\frac{(R-h)}{R} x$ (2) $\frac{(R-h)^2}{R^2} x$ (3) $\frac{(R-h)}{R^2} x$ (4) $\frac{Rx}{(R-h)} x$ (5) $\sqrt{\frac{(R-h)}{R}} x$

- (86) A soap bubble A of radius R is touching the inside of a soap bubble B of radius $2R$. A soap bubble C of radius $3R$ is in external contact with B. The graph that correctly shows the change in pressure along a line through the center of the bubbles is,



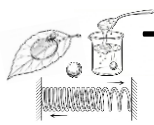
- (87) The variation of pressure with position along the OX axis is correctly shown in,



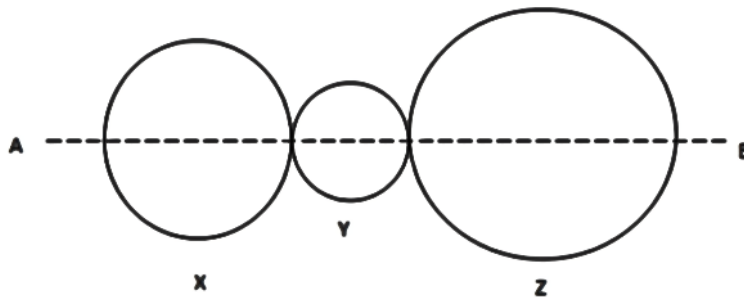
- (88) A flat glass disc of dimensions $10 \text{ cm} \times 8 \text{ cm}$ and thickness 3 mm is placed horizontally with its flat surface resting on a layer of water. If the surface tension of water is $7 \times 10^{-2} \text{ N m}^{-1}$, the additional force exerted by the surface tension on the disc is (Assume that the contact angle between the water and the glass is zero)

(1) $2.52 \times 10^{-2} \text{ N}$ (2) $1.44 \times 10^{-2} \text{ N}$ (3) $1.25 \times 10^{-2} \text{ N}$
(4) $7.26 \times 10^{-2} \text{ N}$ (5) $5.00 \times 10^{-2} \text{ N}$

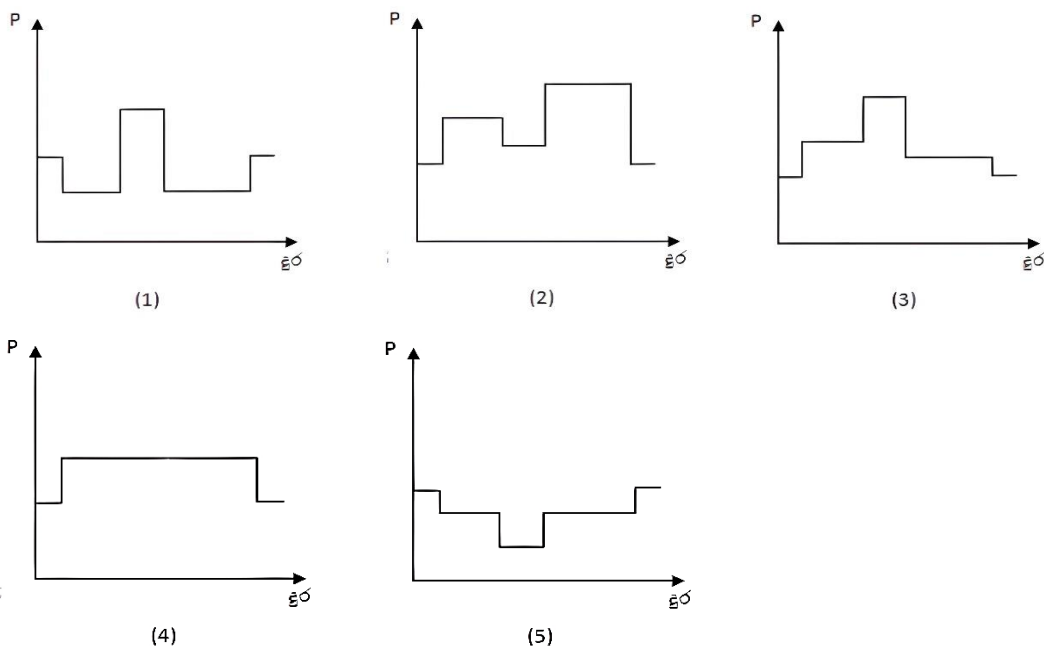




(89)



The three x, y, z soap bubbles are stick together as shown in the image above. The radii of x, y and z vary as $r_y < r_x < r_z$, respectively. The pressure variation from A to B along the line AB is best illustrated by,

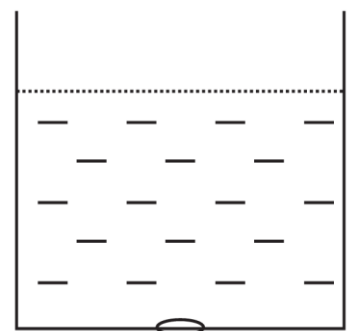


(90) As shown in the figure, a container with a circular hole at the bottom contains a liquid to a height of 5 cm. The surface tension coefficient of the liquid is 0.03 N m^{-1} and the density is 800 kg m^{-3} . Consider the following clauses: (The radius of the hole = 0.5 mm)

- If the excess pressure exerted by the liquid becomes less than the maximum excess pressure that can be maintained within the bubble, the bubble will burst.
- If the radius of the bubble is smaller than or equal to the radius of the hole, the bubble cannot burst.
- In the above case, the liquid does not leak out of the holes

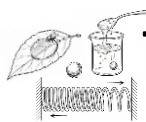
Which of these is/are true?

- a only.
- b only.
- a and b only
- a and c only
- All a, b and c

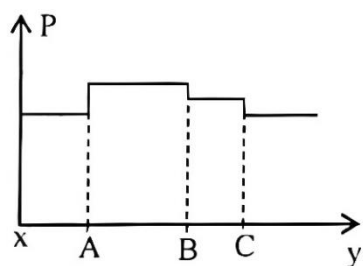
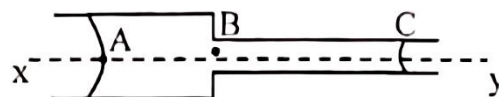


(91) The pressure difference between the external and internal pressures of a single soap bubble is three times that of another soap bubble. The ratio between the volumes of the second and first soap bubbles is,

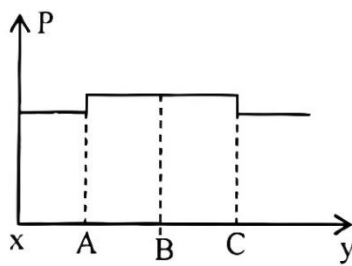
- 3:1
- 1:9
- 27:1
- 1:27
- 1:3



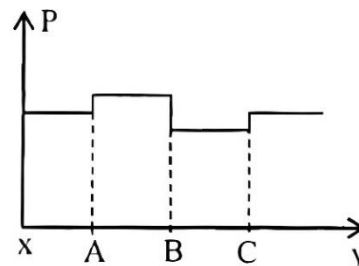
- (92) Two capillary tubes of different cross-sectional radii are connected to each other and holds a water layer. Which graph shows the variation of pressure from x to y?



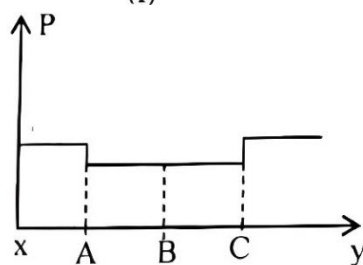
(1)



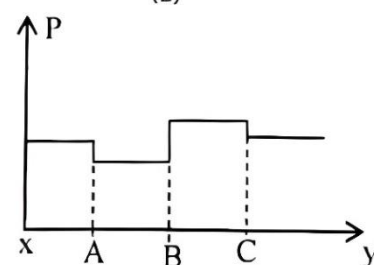
(2)



(3)



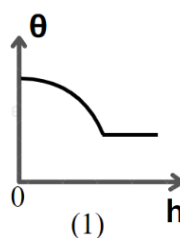
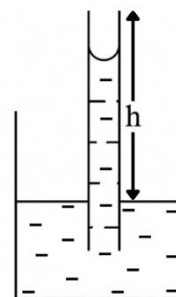
(4)



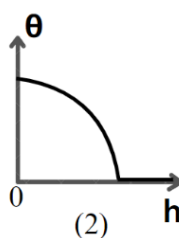
(5)

• **Master Level**

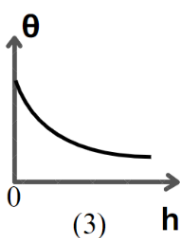
- (93) The image shows a capillary tube immersed in a water beaker. In this case, the contact angle of the water meniscus in the capillary tube with is not zero. Which graph shows the change in the contact angle θ of the water meniscus inside the tube when the vertical height h between the water level in the beaker and the upper end of the capillary tube is changed from zero?



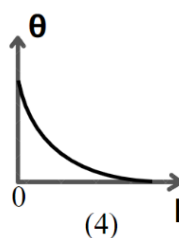
(1)



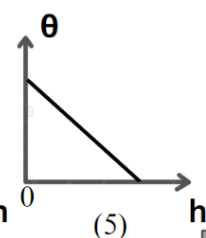
(2)



(3)

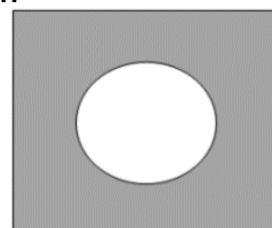


(4)



(5)

- (94) A circular loop made of rubber is placed in a soap film and the center is broken. The surface tension of soap is T and the elastic modulus (force to create a unit extension) of rubber is $50T$,

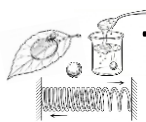


$\frac{\text{Change in diameter}}{\text{Diameter of the loop without soap}} \times 100$ is equal to,

- (1) 0 (2) $2/\pi$ (3) $1/\pi$ (4) $\pi/2$ (5) π

- (95) Water rises to a height h in a glass capillary tube placed vertically in water. The length of the tube in the water is l ($l > h$) and the contact angle between the glass and the water is zero. Now the lower end of the tube is closed in the water, take it out, and the lower end is opened. What is the length of the water column trapped in the tube?

- (1) 0 (2) h (3) $h + l/2$ (4) $h + l$ (5) $2h$



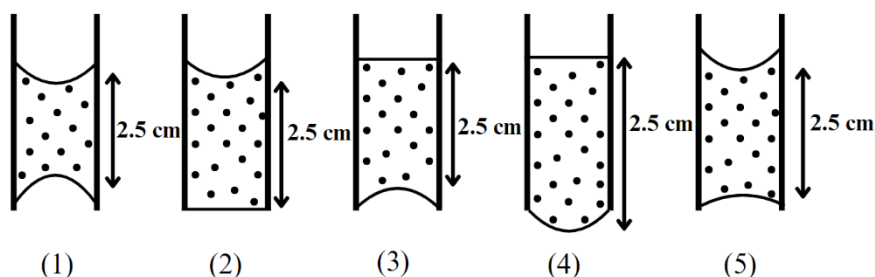
- (96) A capillary tube of inner radius r is vertically immersed in a liquid of density d such that the lower end of the tube is at a depth h from the surface of the liquid. The surface tension of the liquid is T and the contact angle is α . By what amount above atmospheric pressure can the maximum pressure be maintained in the tube by blowing from the upper end?

(1) hdg (2) $hdg + \frac{2T}{r}$ (3) $hdg + \frac{2T \cos \alpha}{r}$
(4) $hdg - \frac{2T}{r}$ (5) $hdg - \frac{2T \cos \alpha}{r}$

- (97) It was found that the capillary rise of water in a capillary tube of inner radius R made of metal is the same as that given by a glass capillary tube of inner radius r . If the contact angle between water and glass is zero, what is the contact angle between water and metal?

(1) 0 (2) $\cos^{-1} \left[\frac{r}{R} \right]$ (3) $\cos^{-1} \left[\frac{R}{r} \right]$
(4) $\cos^{-1} \left[\frac{r}{2R} \right]$ (5) $\cos^{-1} \left[\frac{R}{2r} \right]$

- (98) Water rose 4 cm above the surface of the water outside in a capillary tube partially submerged vertically in a water beaker. When it was carefully pulled out of the water, a slight jerk made a little water inside it to leave, making a 2.5 cm height. The correct representation of the position of the water in the tube when it is placed vertically in air is,



- (99) When eight water droplets of radius r come together to form a single water drop, the surface energy loss becomes entirely kinetic energy. The answer that best represents the maximum velocity attained by the single water drop is, (The surface tension coefficient of water is T and the density of water is ρ)

(1) $\sqrt{\frac{T}{\rho r}}$ (2) $\sqrt{\frac{2T}{\rho r}}$ (3) $\sqrt{\frac{3T}{\rho r}}$
(4) $2\sqrt{\frac{T}{\rho r}}$ (5) $\sqrt{\frac{6T}{\rho r}}$

- (100) A drop of mercury of radius R falling from a height H above the ground breaks up into n number of smaller droplets. If the density of mercury is ρ and assuming that there is no loss of energy as thermal energy or sound energy, the surface tension for mercury can be expressed as,

(1) $\frac{\rho g H R}{4 \left[\frac{1}{n^3} - 1 \right]}$ (2) $\frac{2 \rho g H R}{3 \left[\frac{1}{n^3} - 1 \right]}$ (3) $\frac{\rho g H R}{3 \left[\frac{2}{n^3} - 1 \right]}$
(4) $\frac{2 \rho g H R}{4 \left[\frac{1}{n^3} + 1 \right]}$ (5) $\frac{\rho g H R}{3 \left[\frac{1}{n^3} - 1 \right]}$

MCQ ANSWERS

<i>Elasticity</i>							
(15)	3	(16)	1	(17)	5	(18)	3
(19)	3	(20)	2	(21)	1		
(22)	5	(23)	5	(24)	3	(25)	5
(26)	4						

<i>Viscosity</i>						
(42) 5	(43) 3	(44) 4	(45) 3	(46) 4	(47) 4	(48) 5
(49) 2	(50) 4	(51) 4	(52) 4	(53) 2	(54) 1	(55) 3
(56) 4	(57) 1	(58) 4	(59) 4			

Surface Tension							
(78) 1	(79) 4	(80) 4	(81) 5	(82) 1	(83) 2	(84) 3	
(85) 2	(86) 3	(87) 4	(88) 1	(89) 3	(90) 5	(91) 4	
(92) 4	(93) 1	(94) 2	(95) 5	(96) 2	(97) 3	(98) 5	
(99) 5	(100) 5						

This image shows a full page of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

4